ST G3-Hybrid PLC&RF GUI
Quick start guide

Rev. 9

March 2024

Introduction

- Scope of this presentation is to help you on starting the evaluation of G3-Hybrid communication with the G3 GUI
- Steps to be performed:
  1. Upgrade to the latest FW package related to the development kit
  2. Open the GUI and configure the development kit
  3. Perform communication tests
     1. PHY layer
     2. MAC layer
     3. IPv6 layer
  4. Optional: use one additional kit as sniffer
  5. Non-Volatile Memory operations

Sections

1 PHY communication tests

2 MAC communication tests

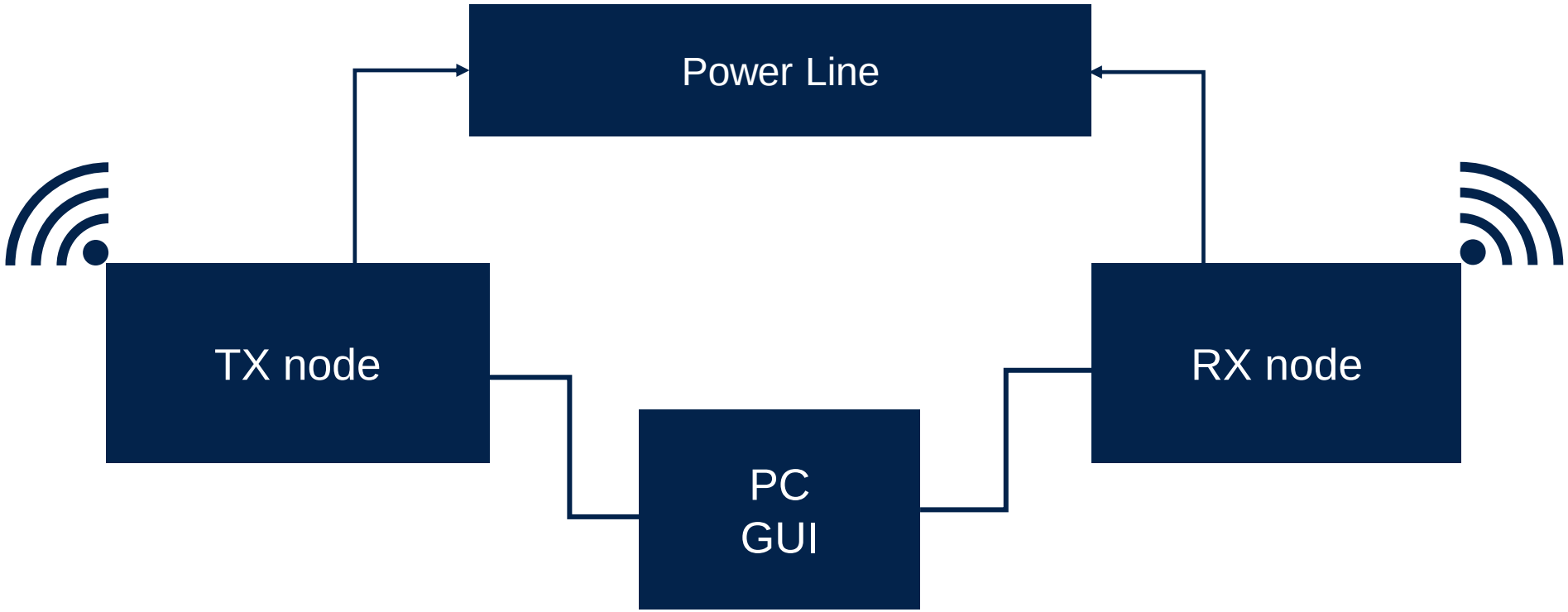
3 UDP-IPv6 communication tests

4 Sniffer mode

5 NVM operations

Basic communication test setup

- Basic setup with two nodes

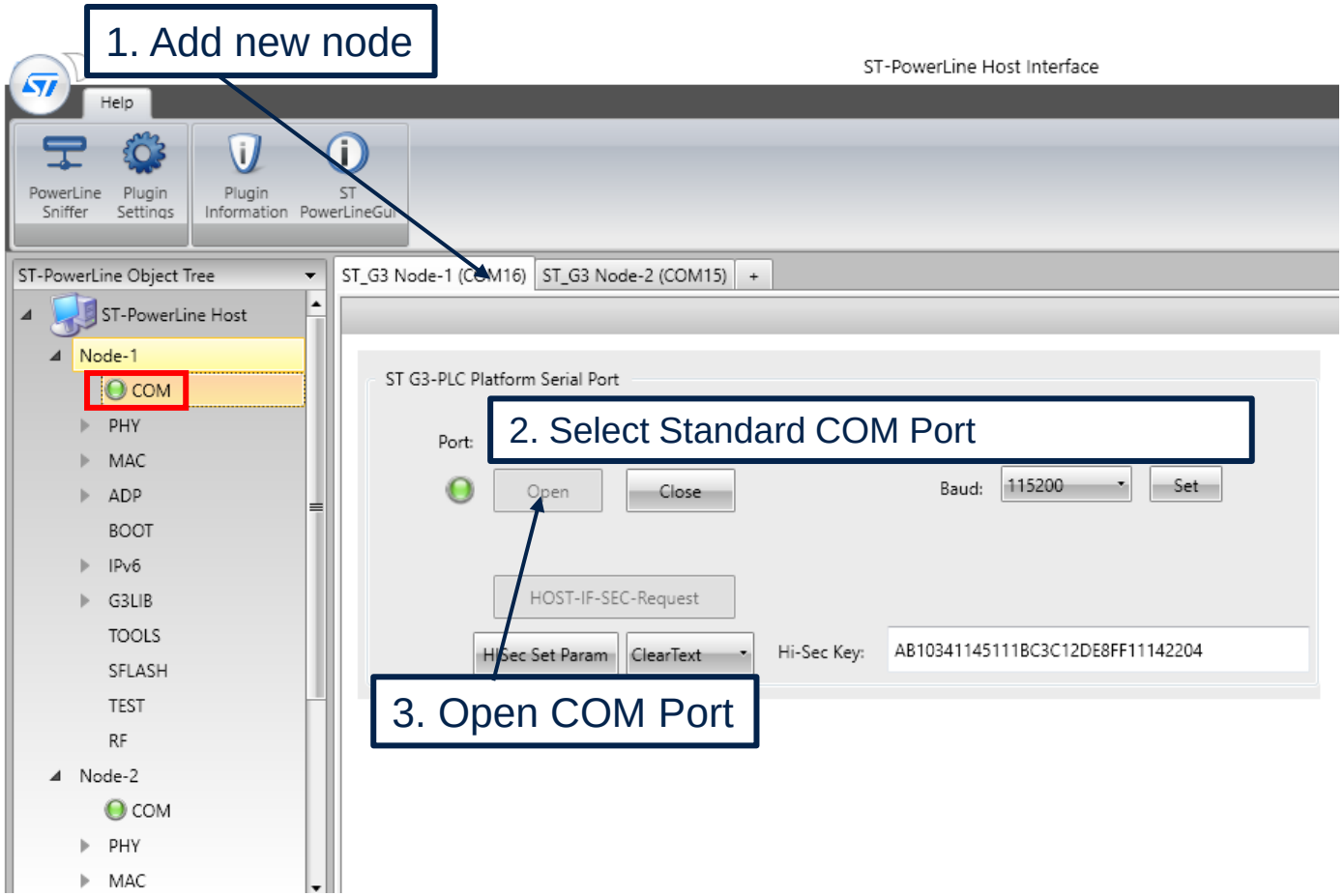


PHY communication tests

Communication setup steps

1. Open G3 GUI: unzip archive and launch *StPowerLineGui.exe* file
2. Add two new nodes and open the UART connection (Standard COM port)
3. Initialize and configure TX and RX nodes
4. Basic TX and RX test
5. Automated data rate TX and RX test with statistics

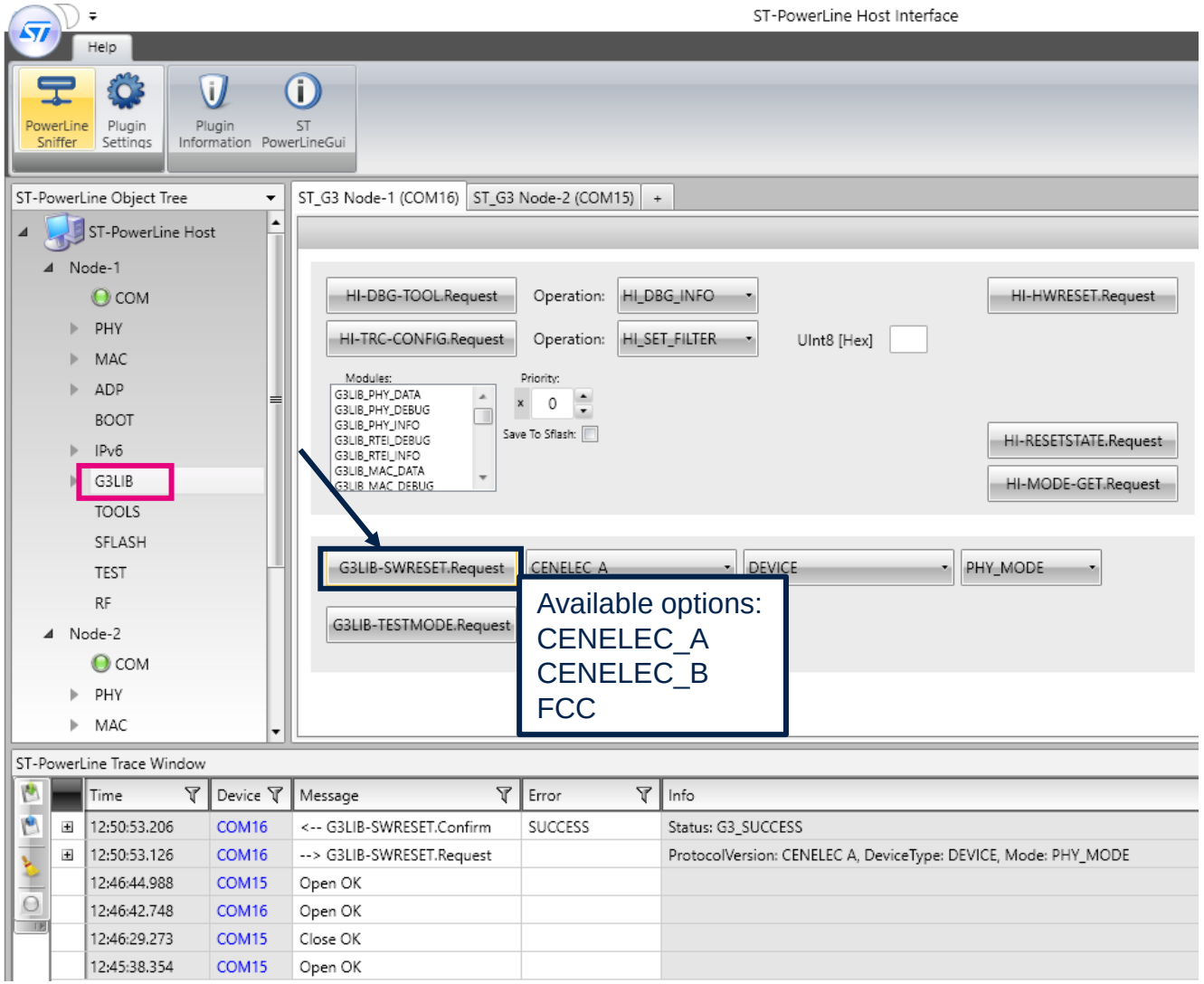
UART connection



Note: you need to create at least 2 nodes for communication test

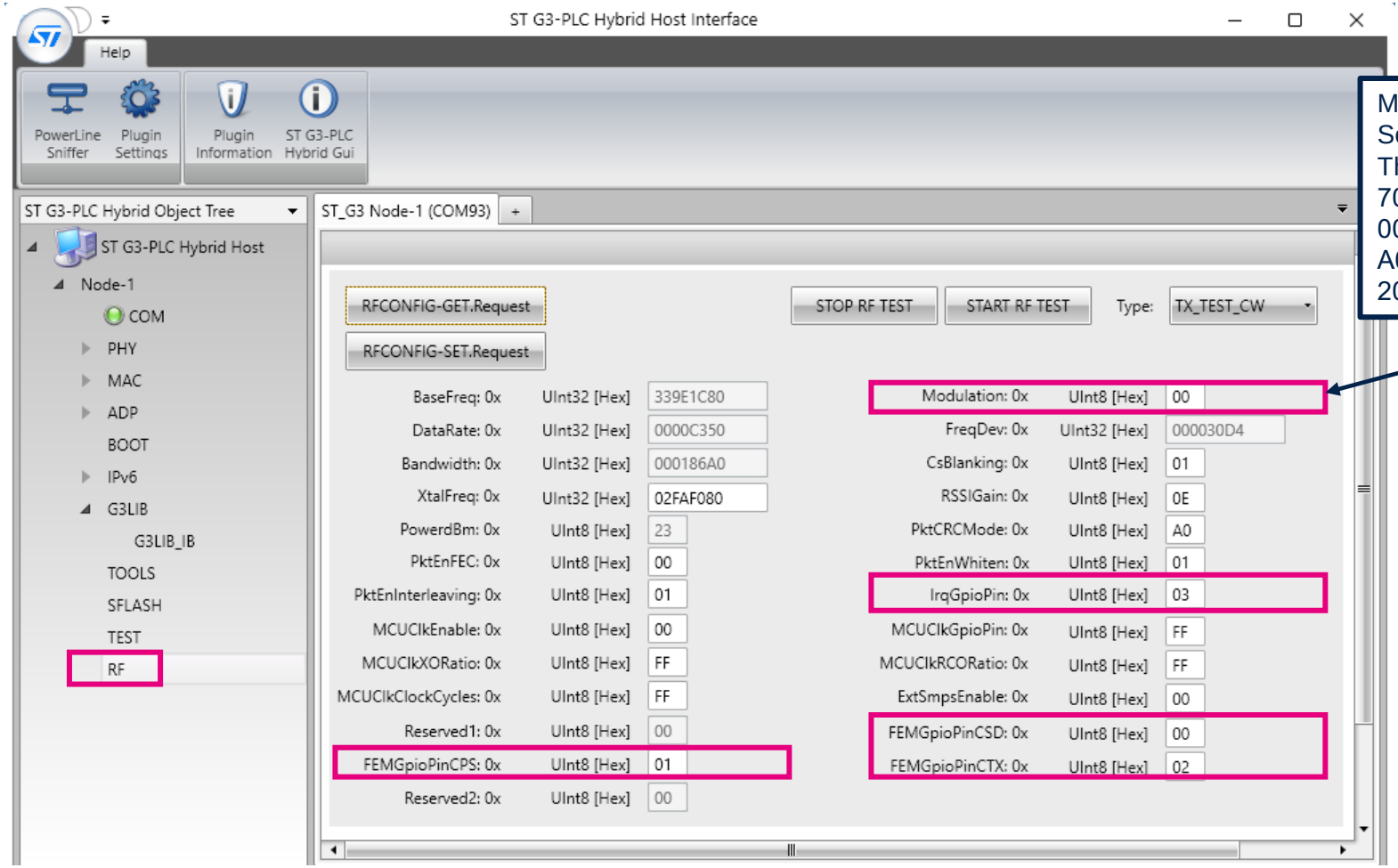
Node initialization

- Repeat the same step for each node



RF configuration

- Repeat the same step for each node



Modulation type:
See ST-G3-PLC_RF_Solutions_AN.pdf
The most common:
70 → No Mod
00 → 2-FSK
A0 → G2FSK with BT = 0.5
20 → G2FSK with BT = 1

RF configuration

- Set RF G3 PIBs for each node

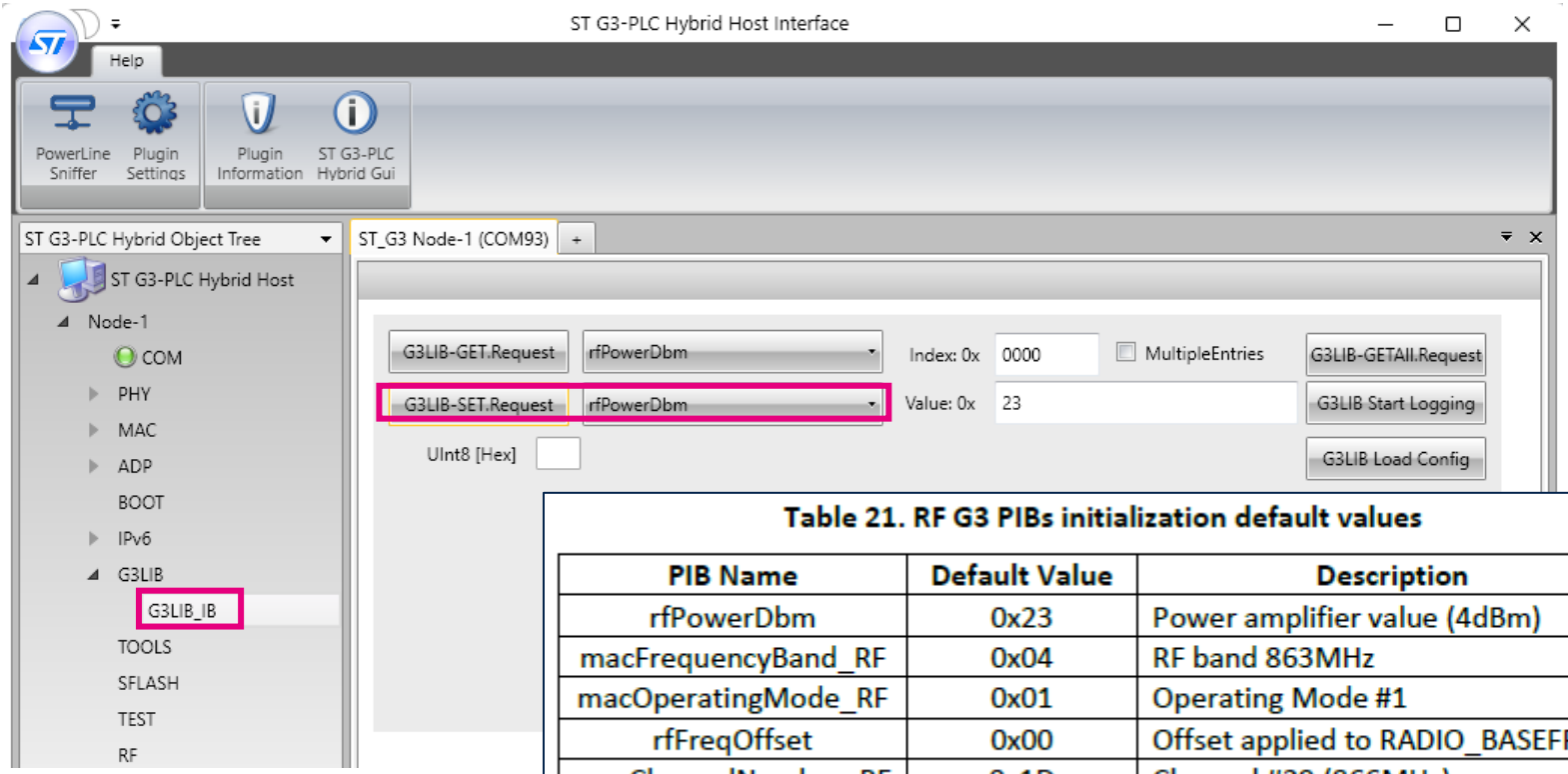


Table 21. RF G3 PIBs initialization default values

PIB Name	Default Value	Description
rfPowerDbm	0x23	Power amplifier value (4dBm)
macFrequencyBand_RF	0x04	RF band 863MHz
macOperatingMode_RF	0x01	Operating Mode #1
rfFreqOffset	0x00	Offset applied to RADIO_BASEFREQ
macChannelNumber_RF	0x1D	Channel #29 (866MHz)

- PIB “macChannelNumber_RF” must always be the last one of the five PIBs to be set

  (see Application Note ST G3-PLC Hybrid Solution → RF device initialization)

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RF configuration

- Set RF FEM GPIO PIBs for each node

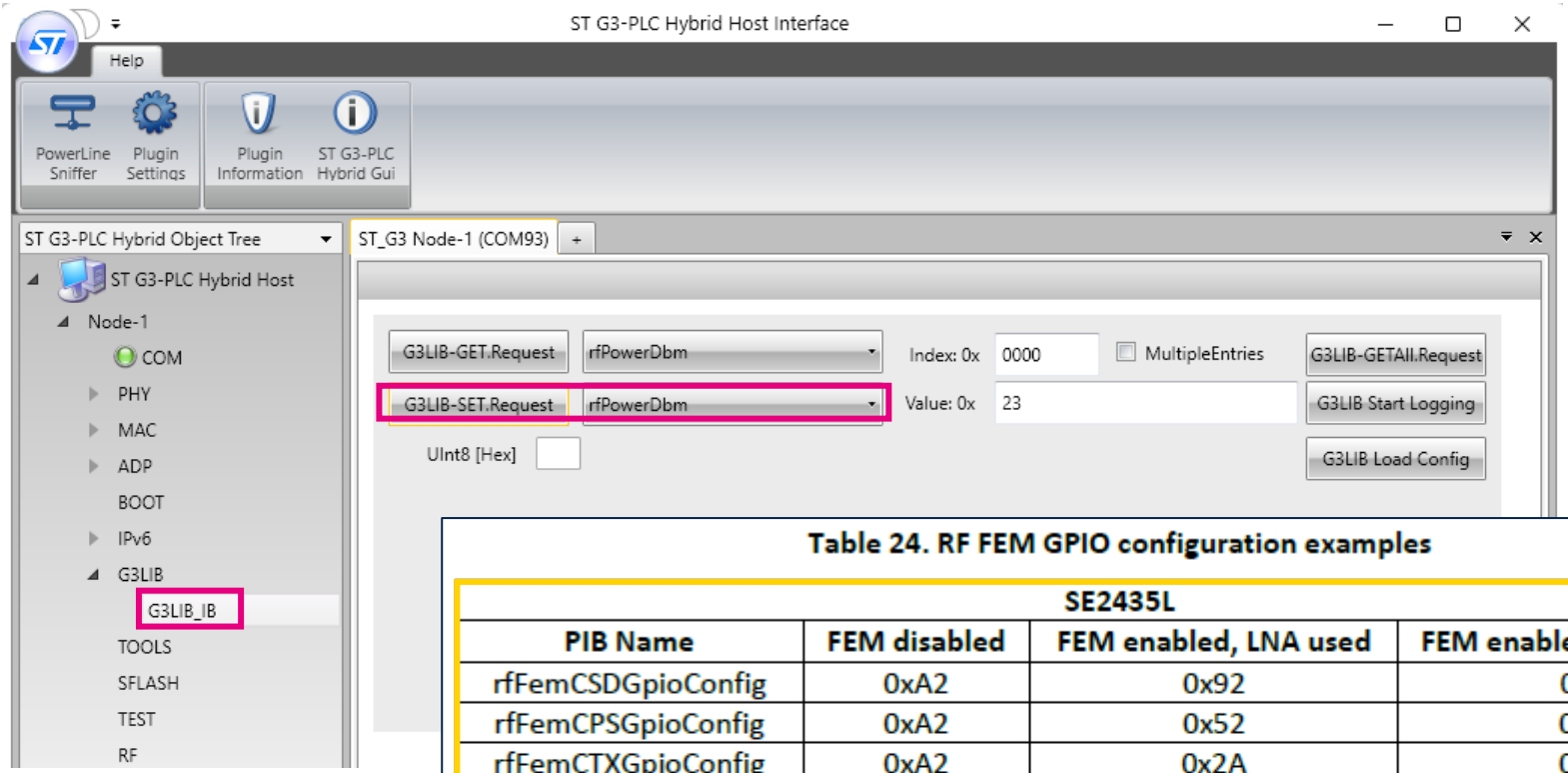
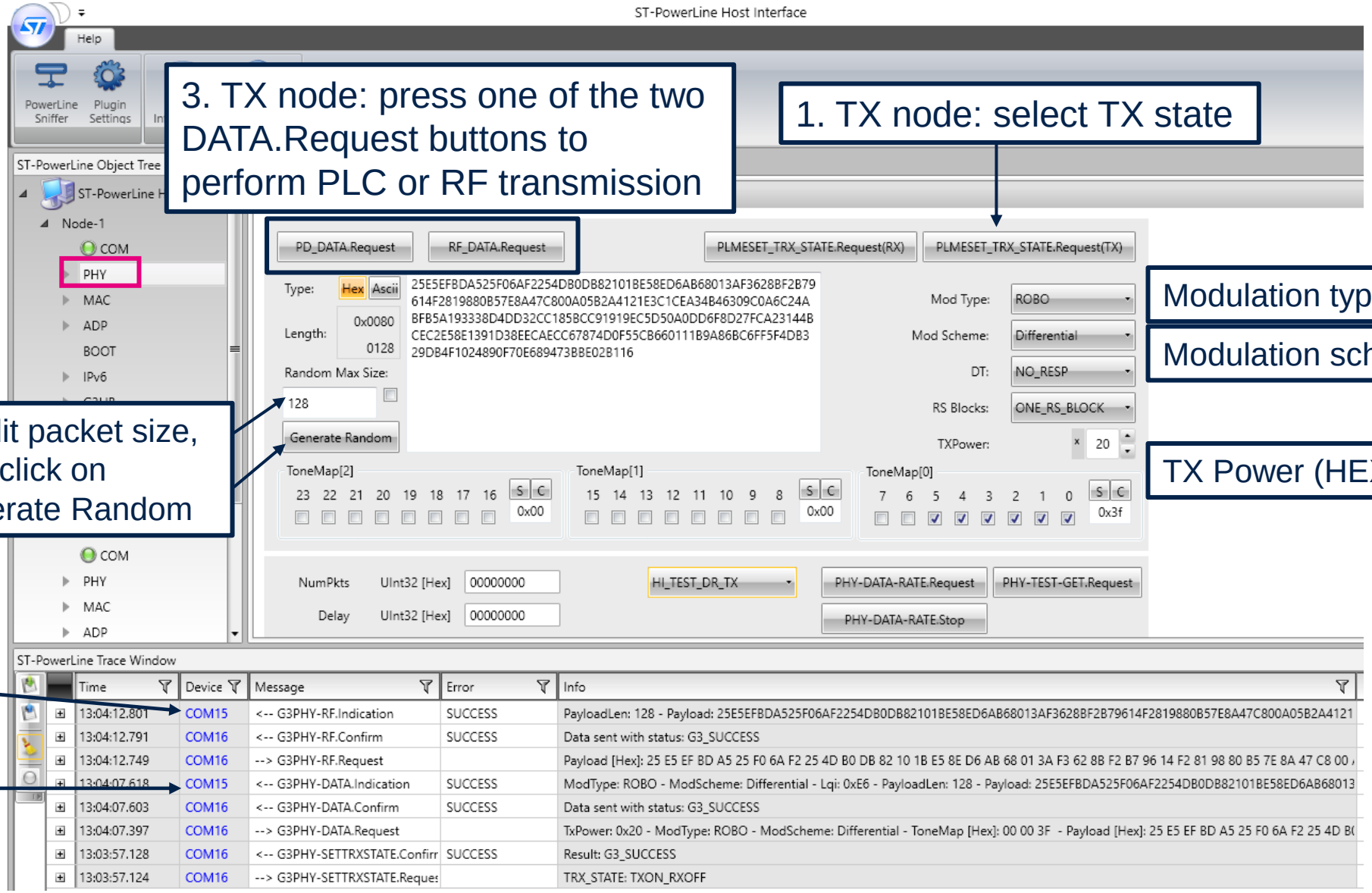


Table 24. RF FEM GPIO configuration examples

SE2435L			
PIB Name	**FEM disabled**	**FEM enabled, LNA used**	**FEM enabled, LNA bypass**
rfFemCSDGpioConfig	0xA2	0x92	0x92
rfFemCPSGpioConfig	0xA2	0x52	0xA2
rfFemCTXGpioConfig	0xA2	0x2A	0x2A
rfFemRxGain	0x00	0x10 (16dB)	0xFE (-2dB)

SKY66423			
PIB Name	**FEM disabled**	**FEM enabled, PA used**	**FEM enabled, PA bypass**
rfFemCSDGpioConfig	0xA2	0x92	0x92
rfFemCPSGpioConfig	0xA2	0x92	0xA2
rfFemCTXGpioConfig	0xA2	0x2A	0x2A
rfFemRxGain	0x00	0x12 (18 dB)	0x12 (18 dB)

PHY TX settings and basic test



3. TX node: press one of the two DATA.Request buttons to perform PLC or RF transmission

1. TX node: select TX state

Modulation type

Modulation scheme

TX Power (HEX)

2. Edit packet size, then click on Generate Random

RF DATA.Indication from RX node

PLC DATA.Indication from RX node

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PHY DATA.Indication from RX node

- PLC PHY DATA.Indication details

13:04:07.618 | COM15 | <-- G3PHY-DATA.Indication | SUCCESS | ModType: ROBO - ModScheme: Differential - Lqi: 0xE6 - PayloadLen: 128 - Payload: 25E5EFBDA525F06AF2254DB0DB82101BE58ED6AB68013

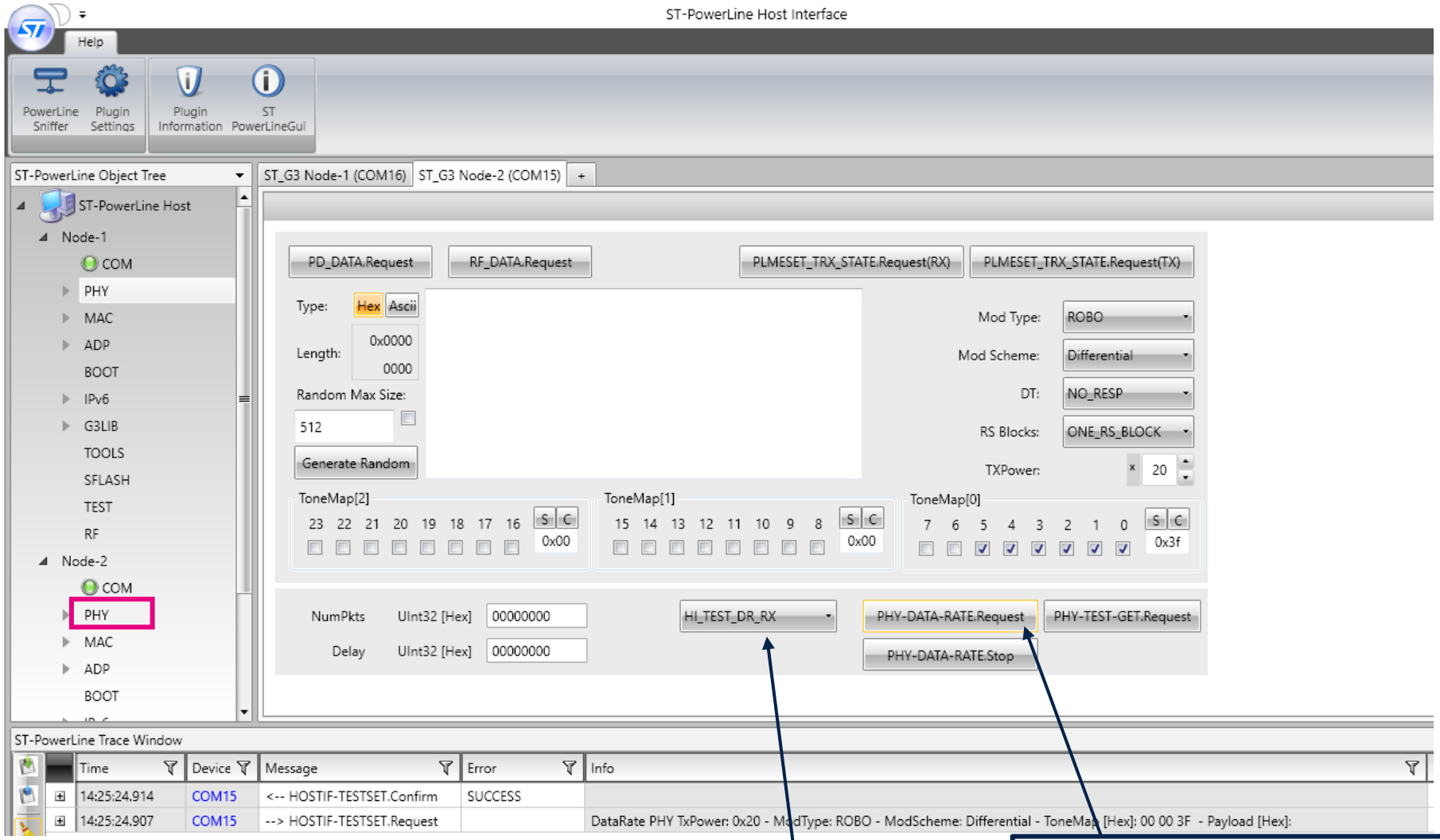
Field Name	Info
ModType:	ROBO
ModScheme:	Differential
ToneMap [Hex]:	00 00 3F
DelimiterType:	NO_RESP
RSBlocks:	ONE_RS_BLOCK
LQI:	0xE6
NoisePower [Hex]:	46 45 45 45 45 45 45 45 45 44 44 44 45 47 47 47 47 47 47 47 46 47 47 46 47 46 46 46 46 46 46 46 46 46 46 46 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
SignalPower [Hex]:	75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 75 76 75 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
PayloadLen:	128
Payload [Hex]:	25 E5 EF BD A5 25 F0 6A F2 25 4D B0 DB 82 10 1B E5 8E D6 AB 68 01 3A F3 62 8B F2 B7 96 14 F2 81 98 80 B5 7E 8A 47 C8 00 A0 5B 2A 41 21 E3 C1 CE A3 4B 46 30 9C 0A 6C 24 AE
HexDump	16 16 42 64 01 00 00 00 00 00 00 3B 3B 3B 3B 3B 3B 3B 3B 3B 3B 3B 3B 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 3A 3A 3A 3A 3A 3A 00 00 00 00 00 00 00 00 00 00 0

- RF PHY DATA.Indication details

13:04:12.801 | COM15 | <-- G3PHY-RF.Indication | SUCCESS | PayloadLen: 128 - Payload: 25E5EFBDA525F06AF2254DB0DB82101BE58ED6AB68013AF3628BF2B79614F2819880B57E8A47C800A05B2A4121

Field Name	Info
RSSI:	0x75
Noise:	0x1C
PayloadLen:	128
Payload [Hex]:	25 E5 EF BD A5 25 F0 6A F2 25 4D B0 DB 82 10 1B E5 8E D6 AB 68 01 3A F3 62 8B F2 B7 96 14 F2 81 98 80 B5 7E 8A 47 C8 00 A0 5B 2A 41 21 E3 C1 CE A3 4B 46 30 9C 0A 6C 24 AE
HexDump	16 16 48 85 00 00 00 00 00 00 00 75 1C 80 00 25 E5 EF BD A5 25 F0 6A F2 25 4D B0 DB 82 10 1B E5 8E D6 AB 68 01 3A F3 62 8B F2 B7 96 14 F2 81 98 80 B5 7E 8A 47 C8 00 A0 5B

PHY Automated data rate RX start

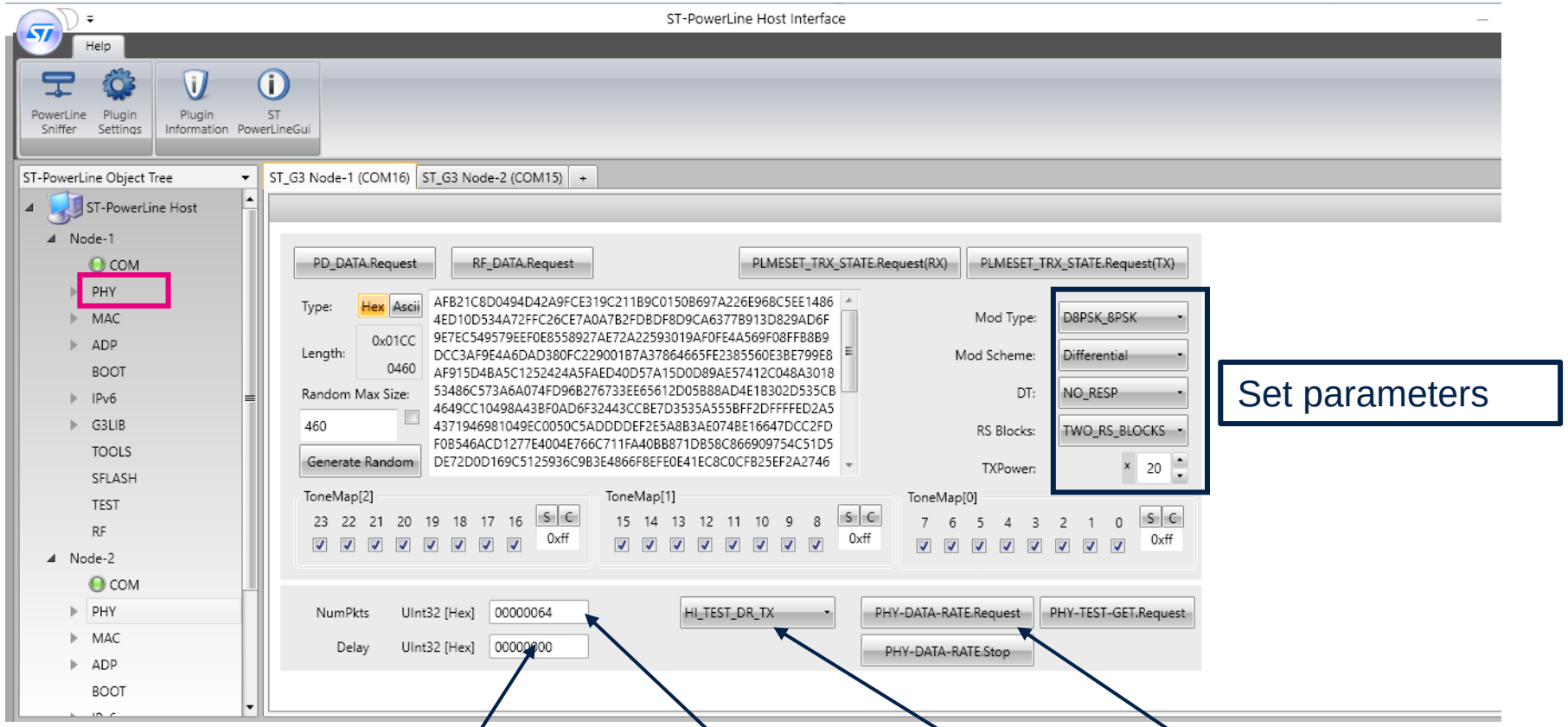


RX test mode

Start RX test: waiting for TX start

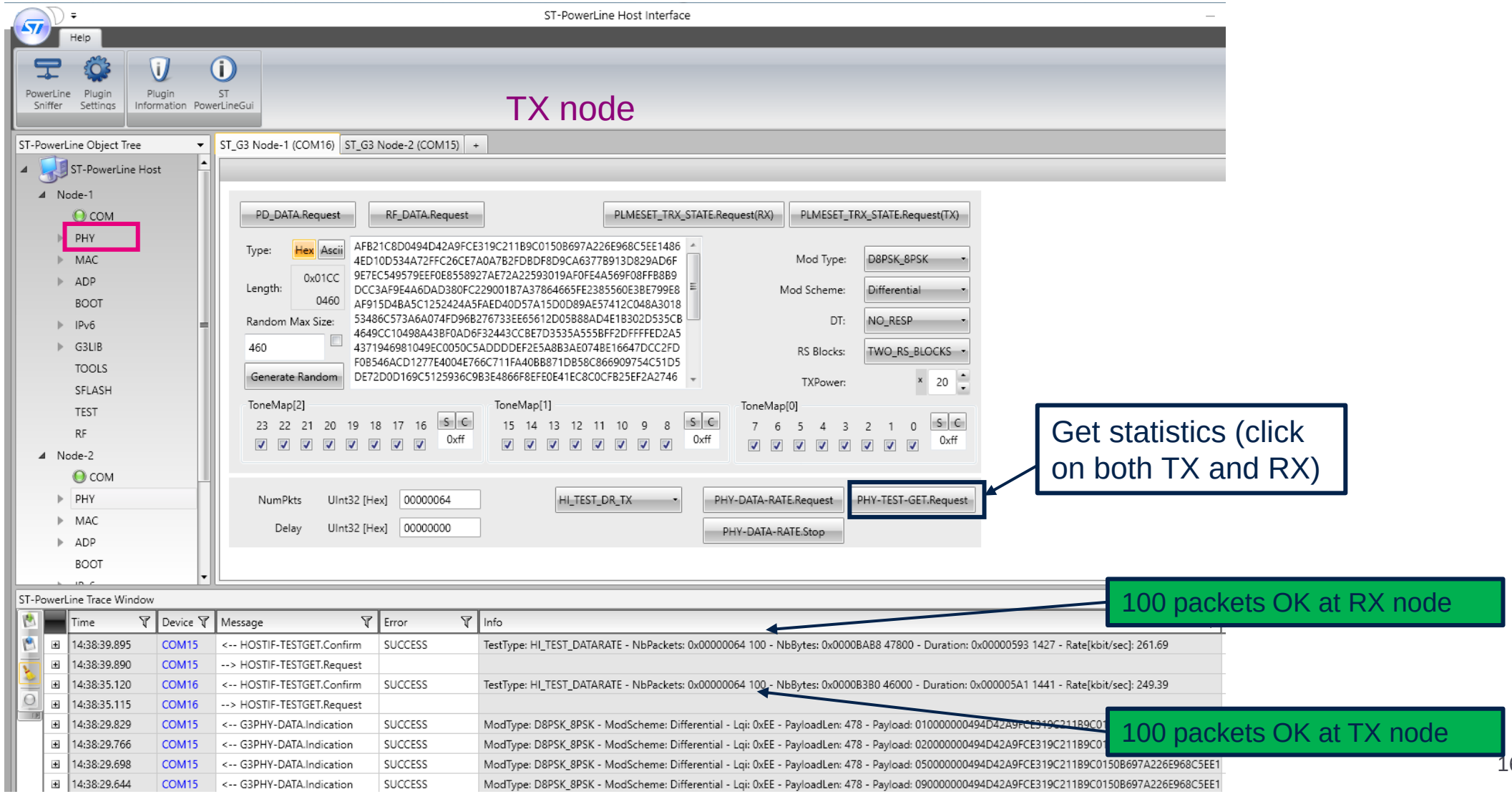
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PHY Automated data rate TX test



PHY Statistics

- Once the transmission has finished, you can see the statistics related to TX and RX packets



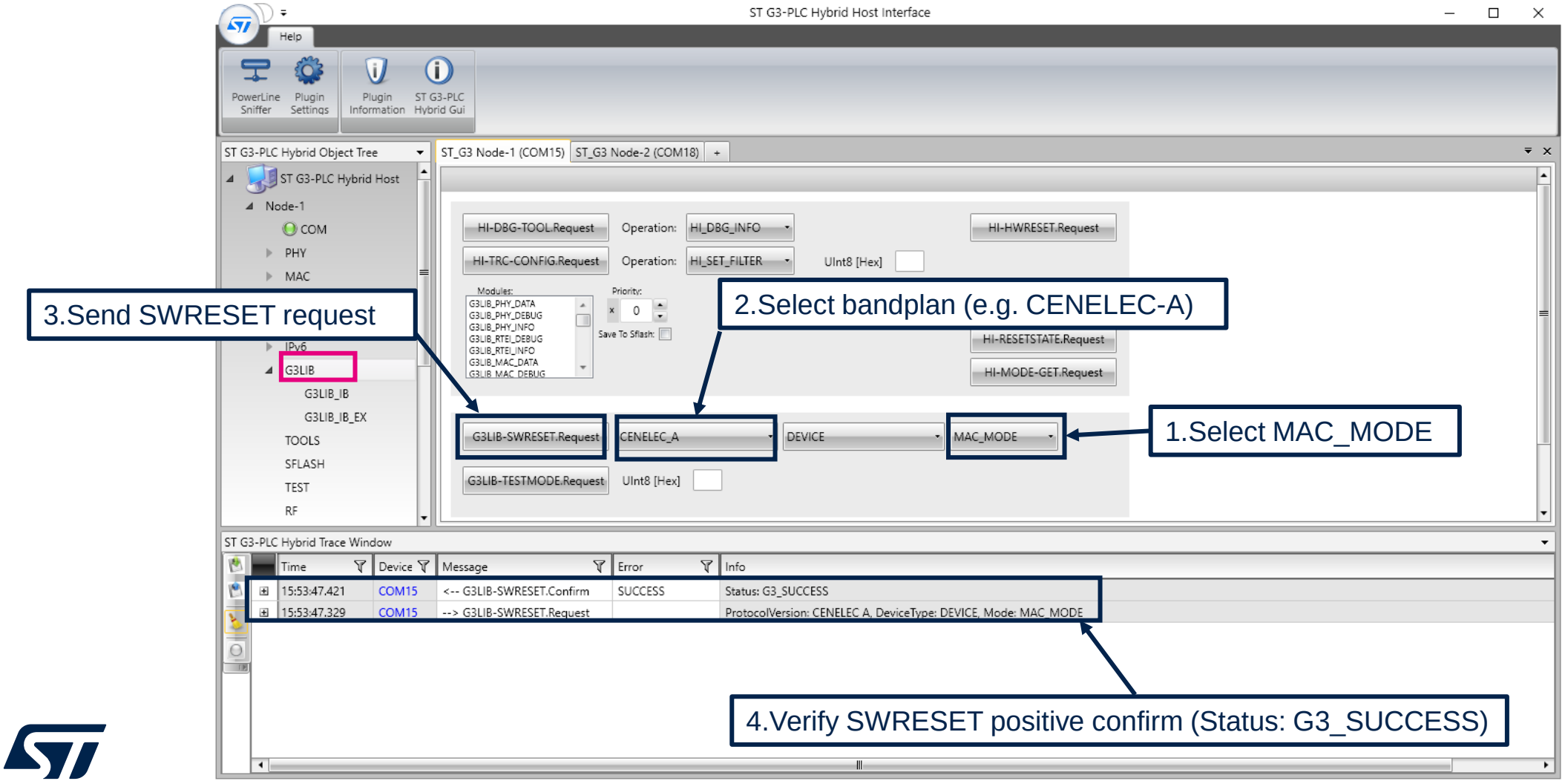
MAC communication tests

Communication setup steps

1. Open G3 GUI: unzip archive and launch *StPowerLineGui.exe* file
2. Add two new nodes and open the UART connection (Standard COM port)
3. Initialize and configure TX and RX nodes
4. Set both nodes in MAC_MODE and select bandplan
5. Check MAC address on both nodes
6. Set and verify PAN ID on both nodes
7. Basic TX and RX test
8. Automated data rate TX and RX test with statistics

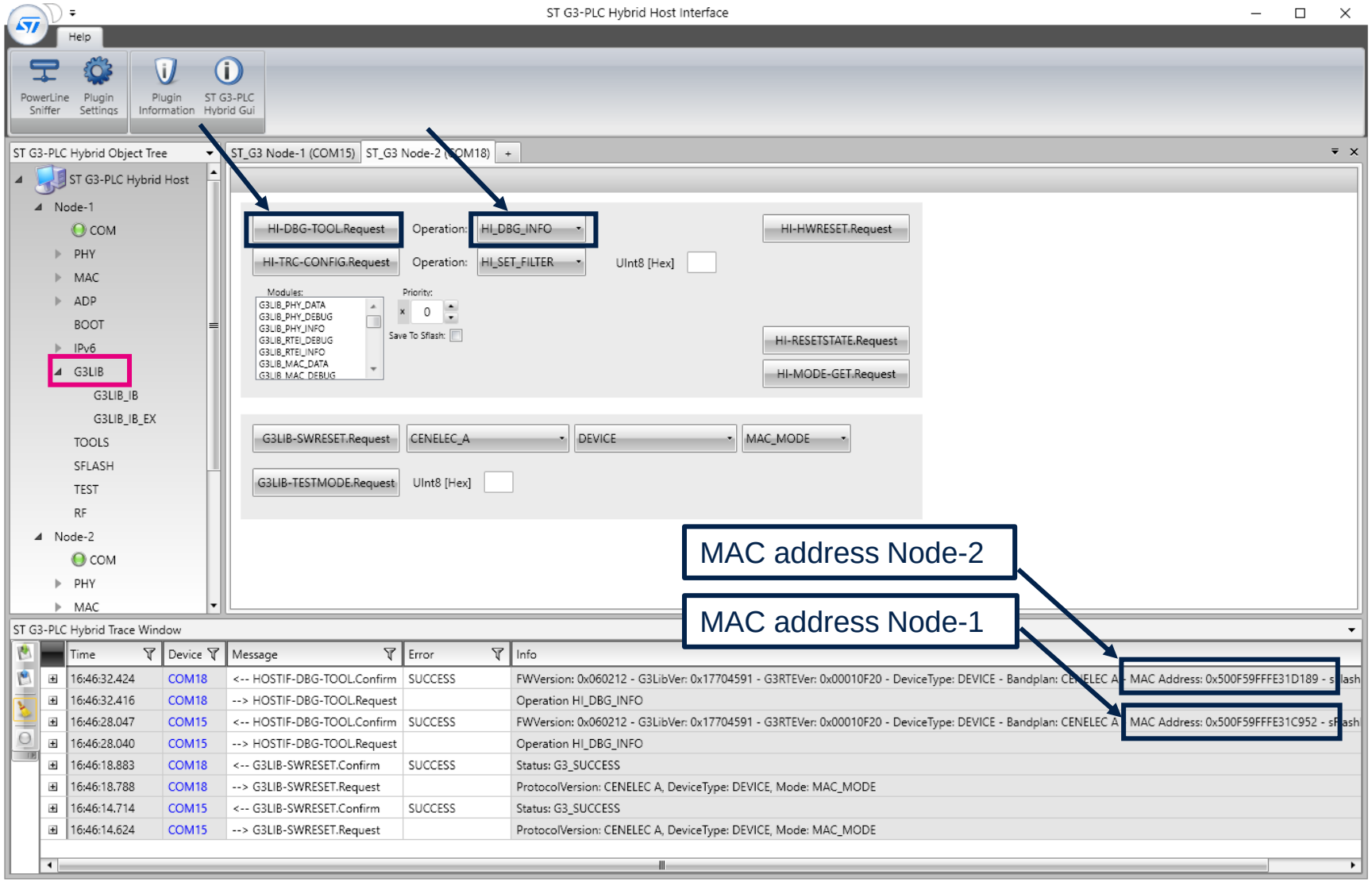
MAC mode and bandplan configuration

- Repeat the following step for each node



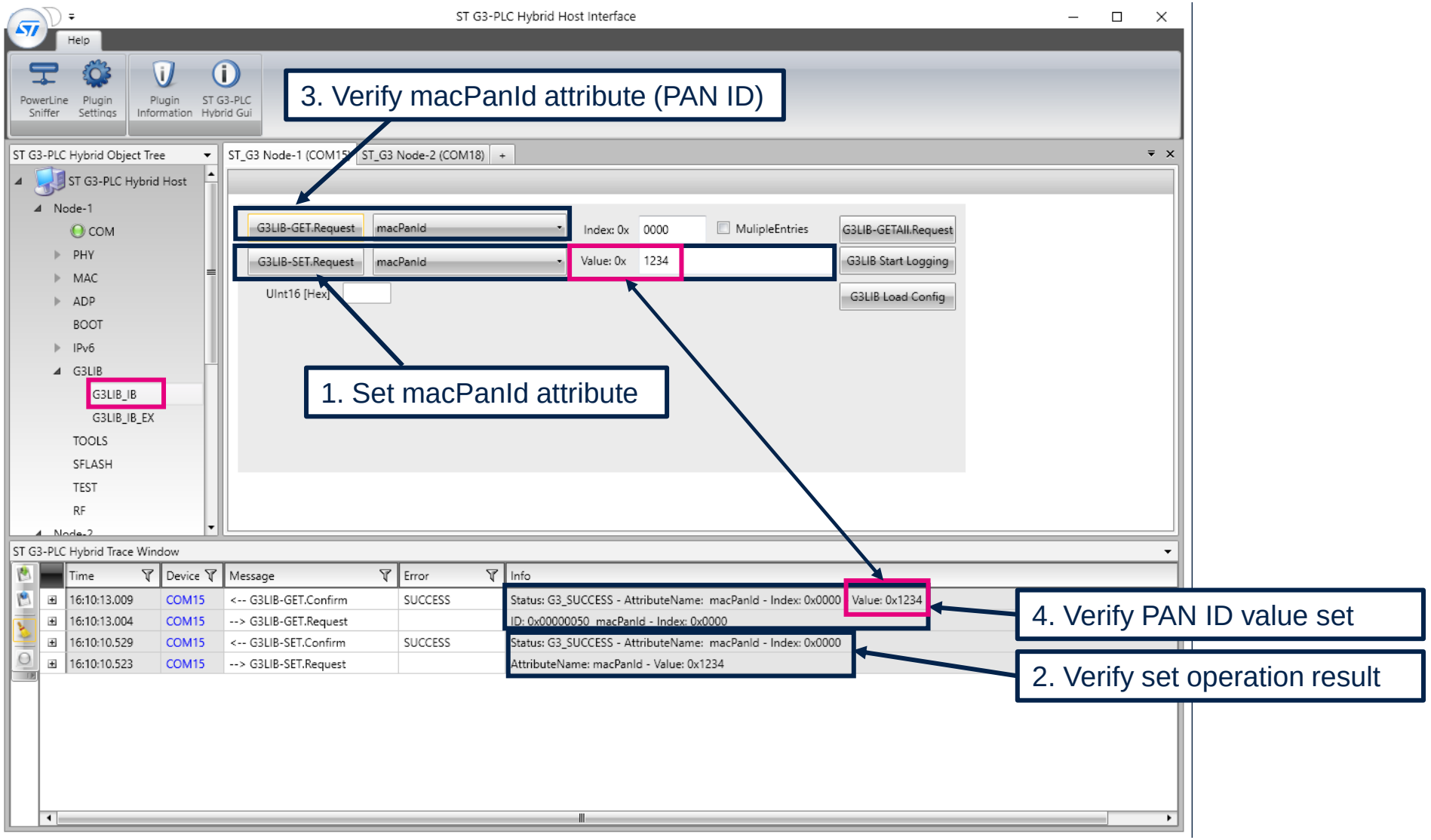
Check MAC address

- Send HI-DBG-TOOL request to each node to get its MAC address

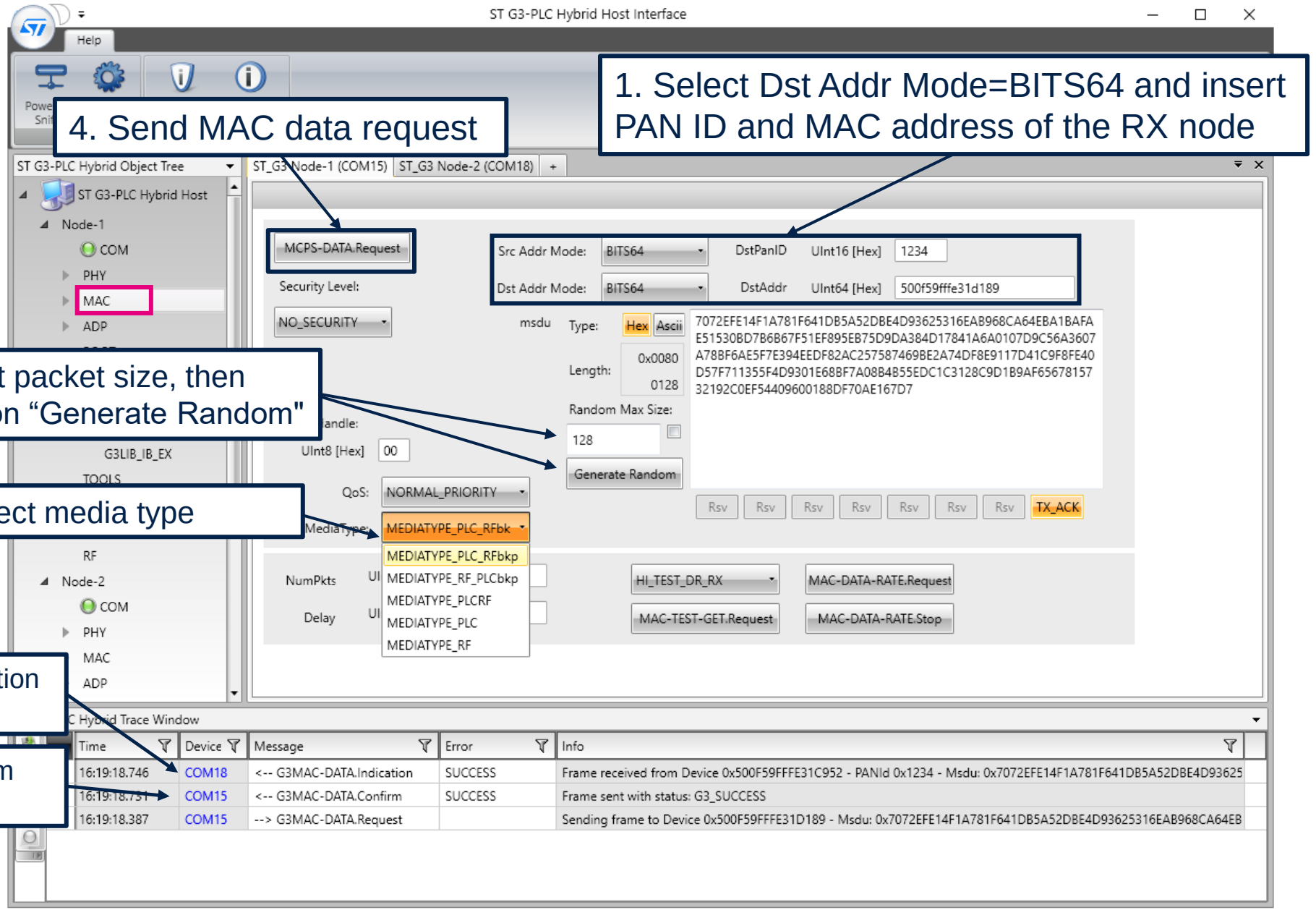


Set and verify PAN ID

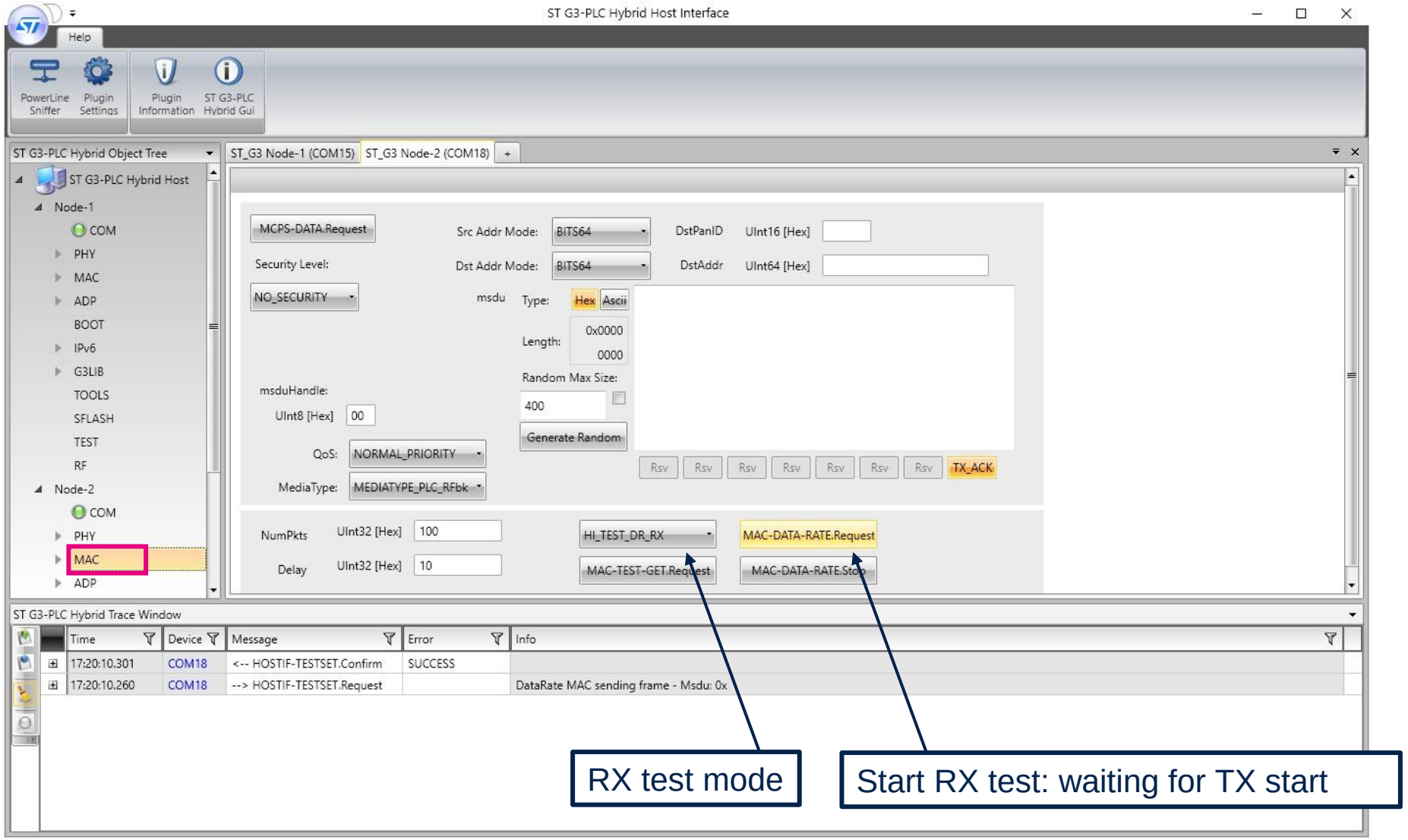
- Repeat the following step with the same PAN ID value for each node



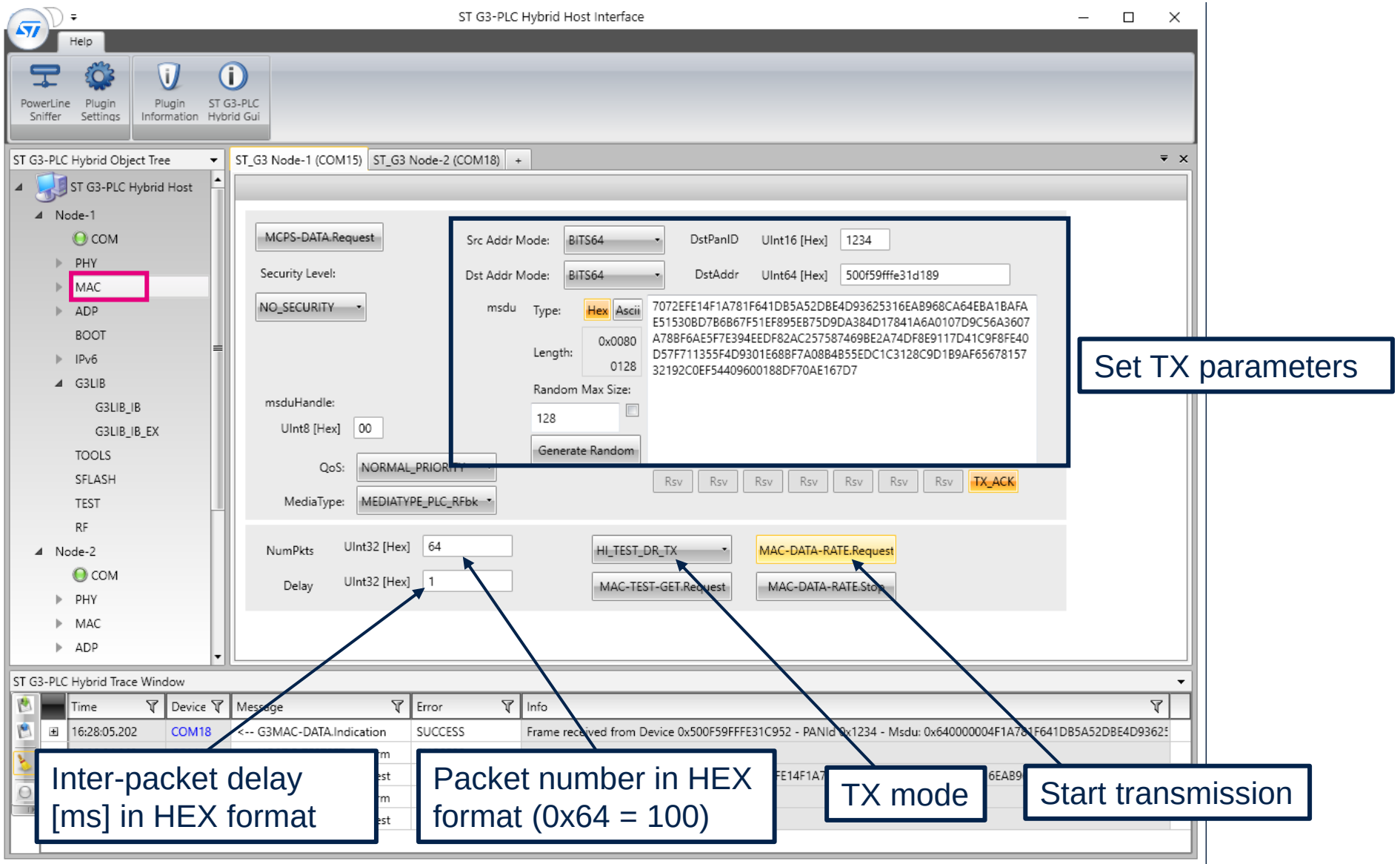
MAC TX setup and basic test



MAC automated data rate RX start

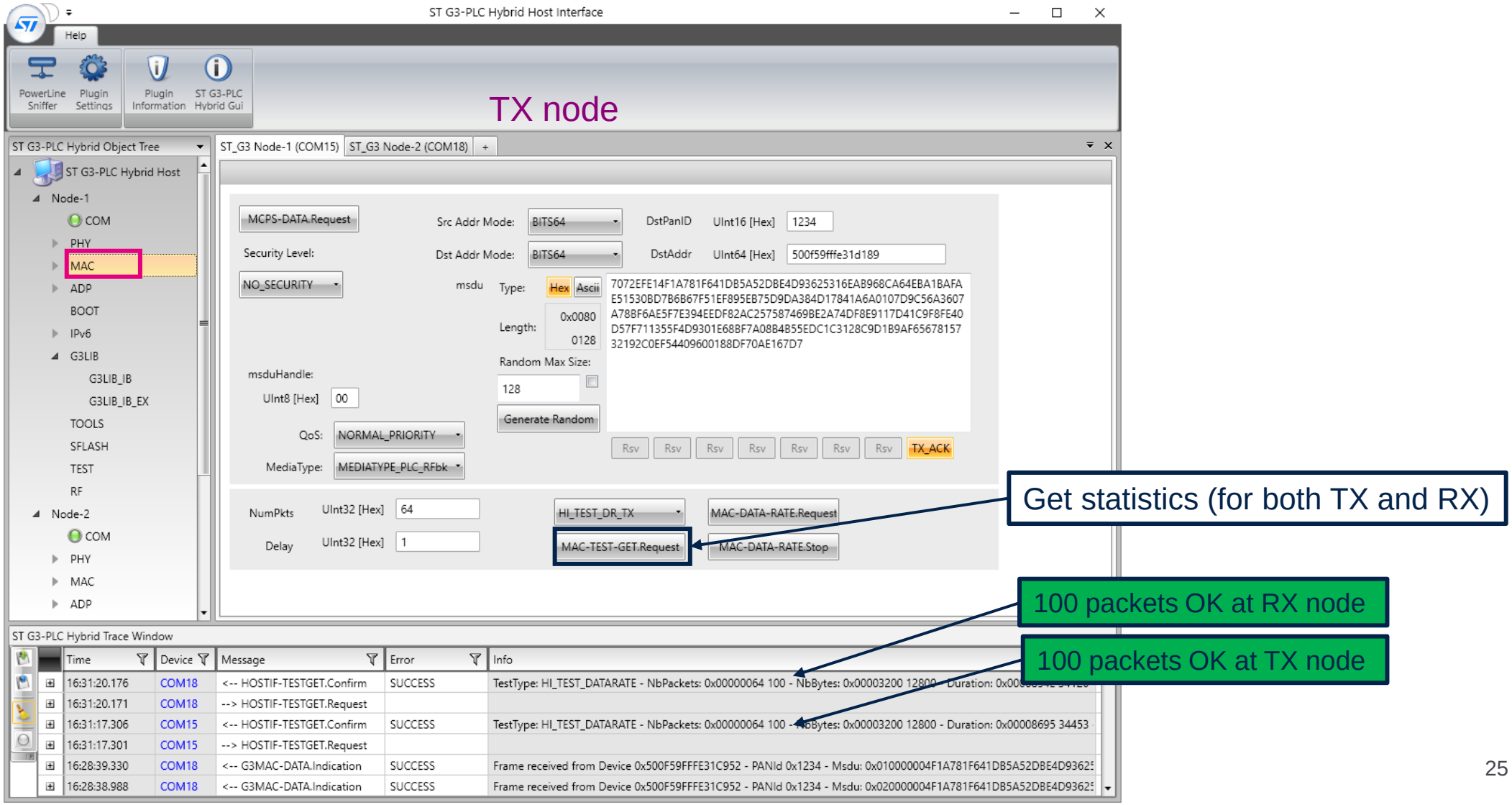


MAC automated data rate TX test



MAC Statistics

- Once the test is complete, it is possible to view the TX and RX statistics

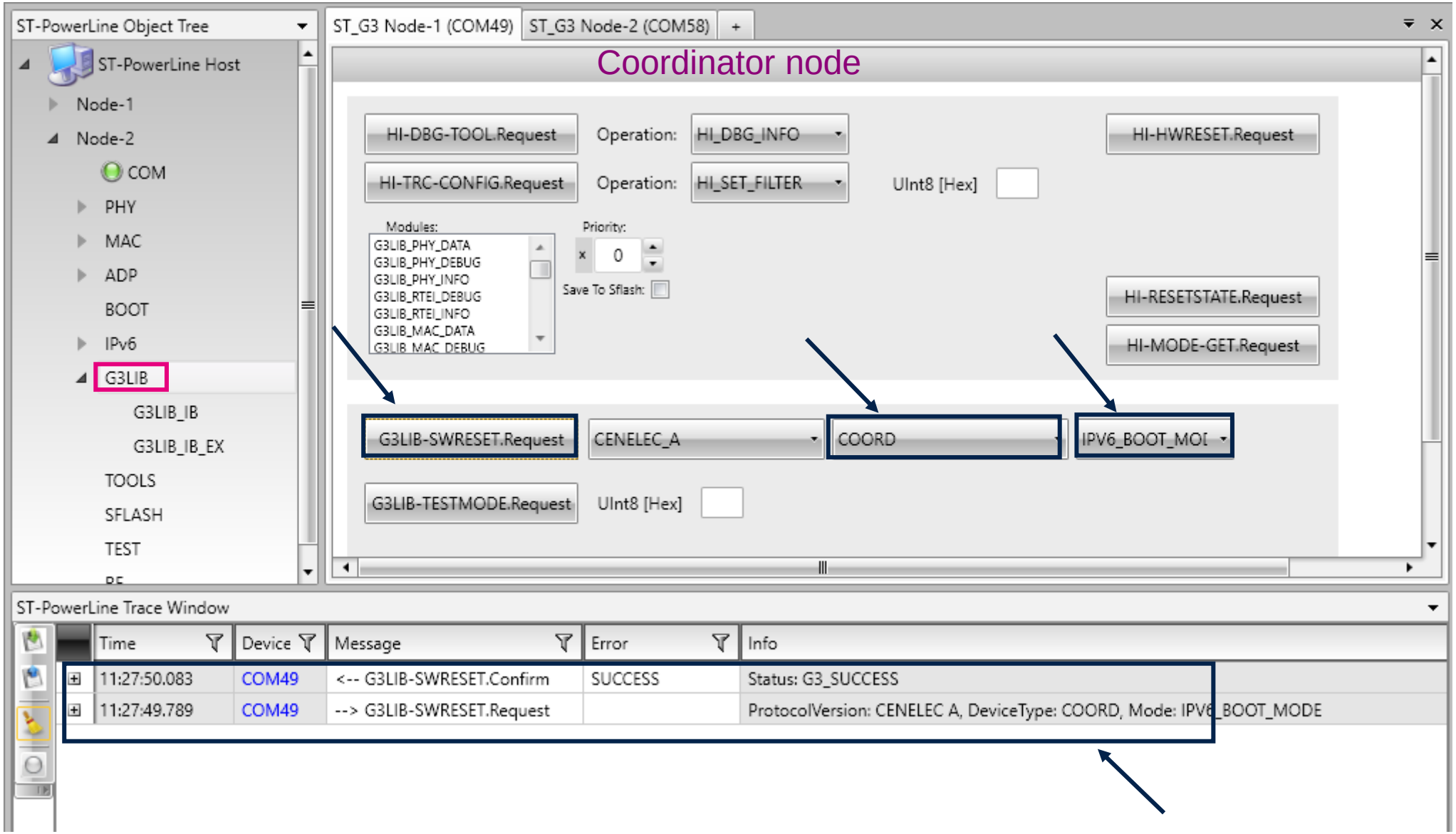


UDP-IPv6 communication tests

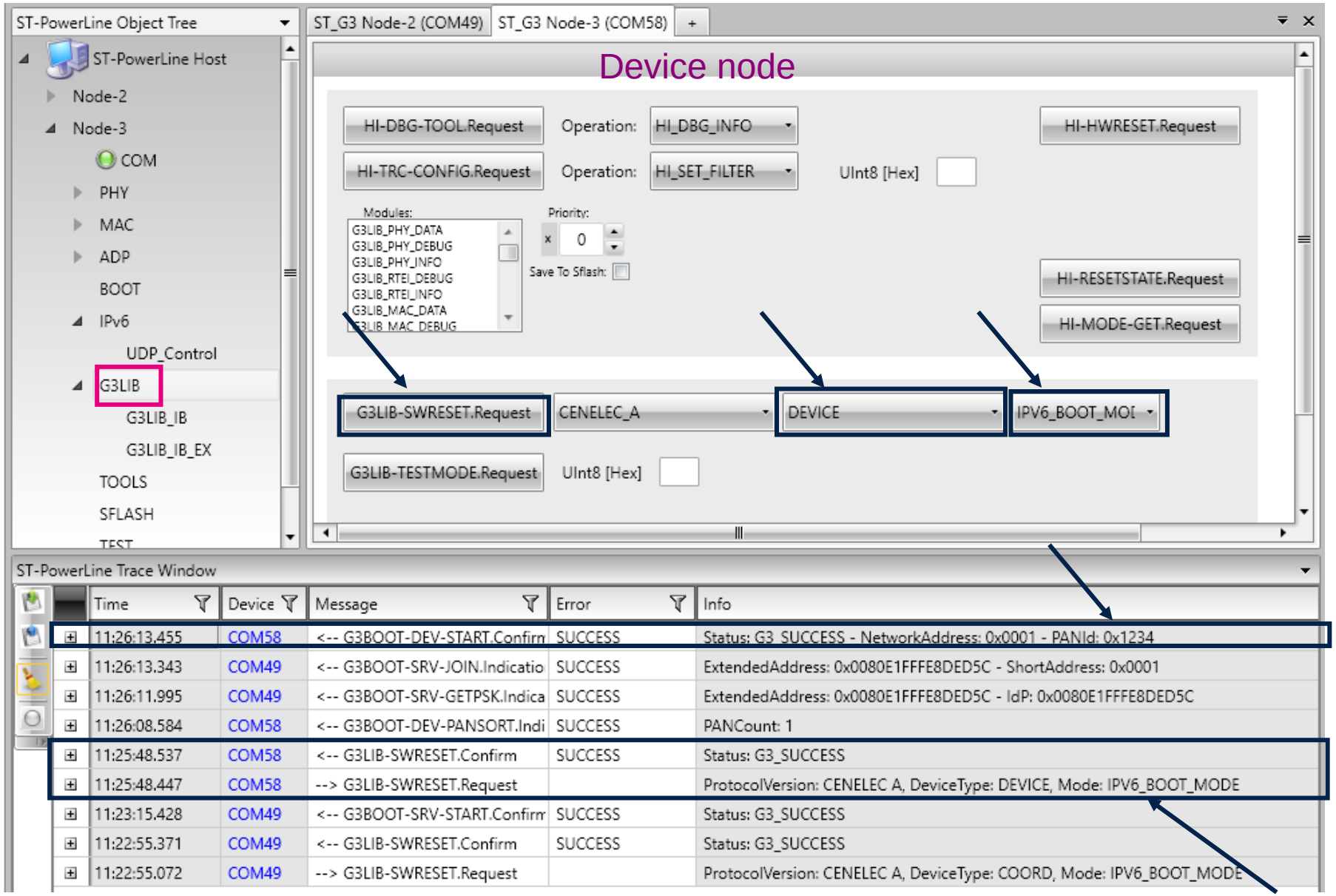
Communication setup steps

1. Open G3 GUI: unzip archive and launch *StPowerLineGui.exe* file
2. Add two new nodes and open the UART connection (Standard COM port)
3. Initialize and configure TX and RX nodes
4. Set one node as COORD and the other as DEVICE in IPV6_BOOT_MODE.
   (*configure the DEVICE one minute after the COORD for faster join procedure)*
5. *Check COORD Short Address*
5. Wait for the bootstrap procedure to complete
6. Get the PAN ID and the Short Address of the DEVICE from the G3BOOT-DEV-START.Confirm
7. Configure UDP unicast connection for both COORD and DEVICE
8. Test communication from COORD to DEVICE
9. Test communication from DEVICE to COORD

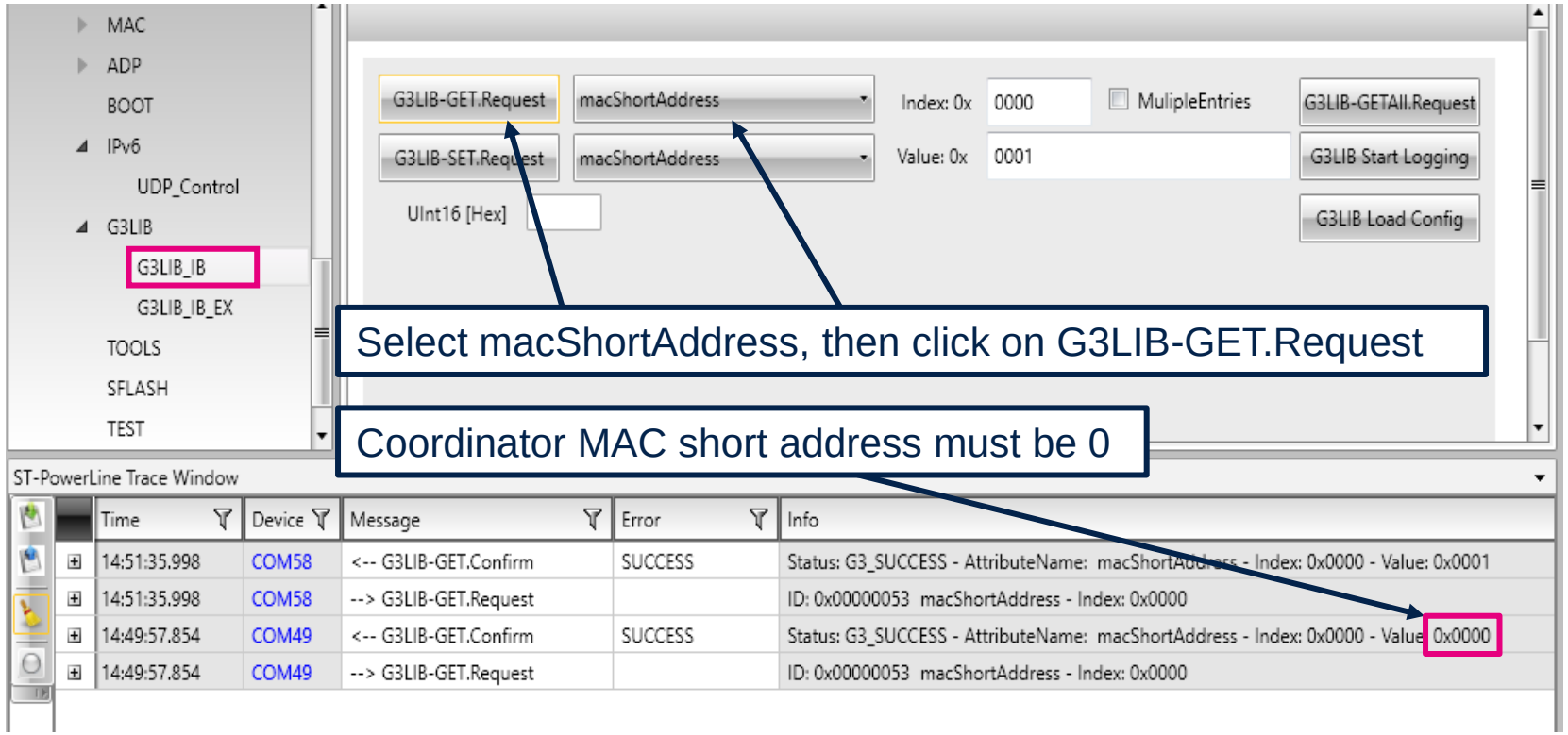
IPv6 boot mode configuration: Coordinator



IPv6 boot mode configuration: Device

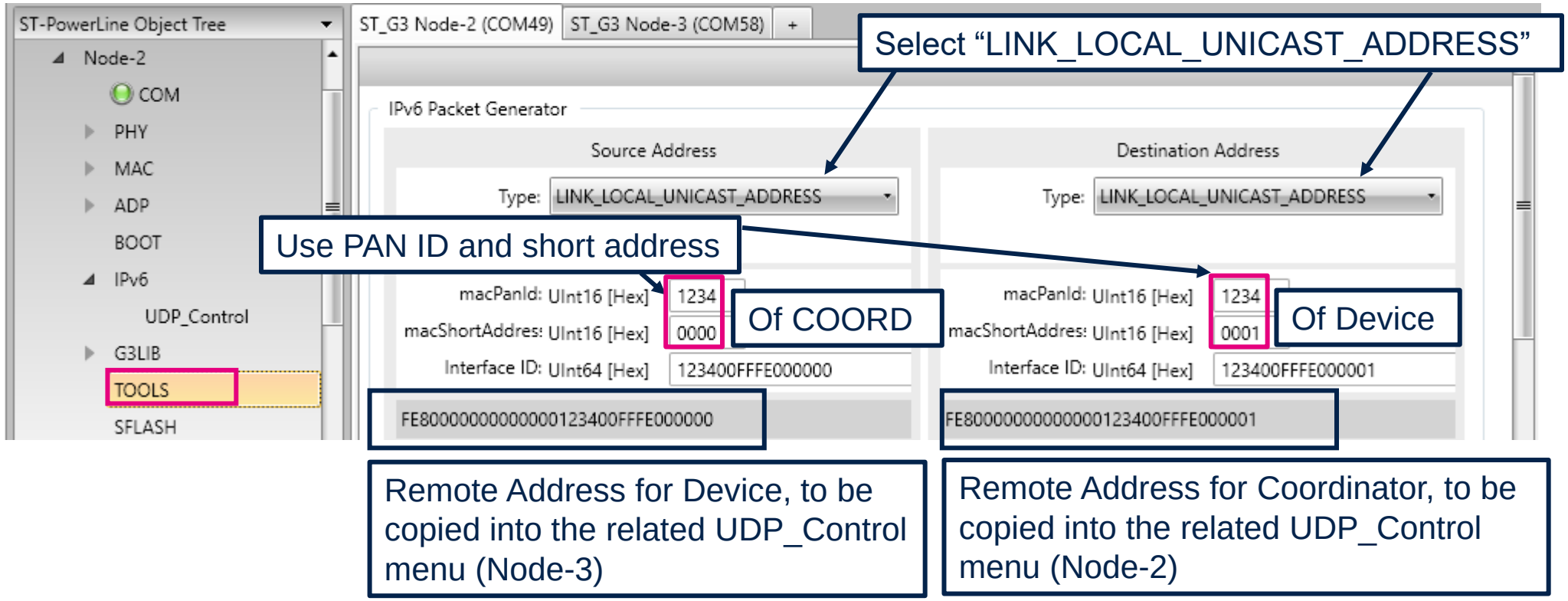


Check MAC short address (Coordinator side)



Unicast connection settings: Coordinator

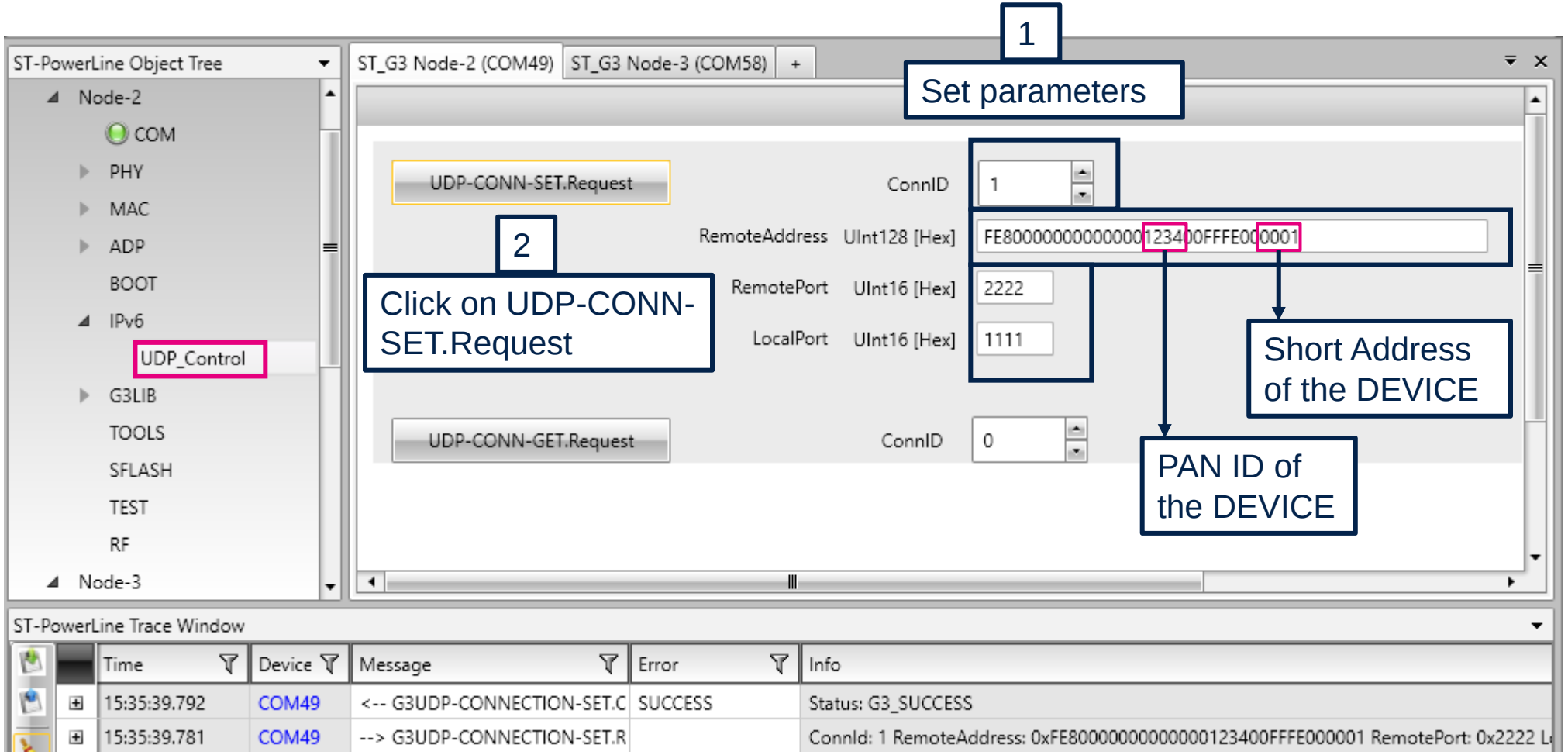
Coordinator node



Note: the PAN ID and the Short Address are embedded in the IPv6 address of the device

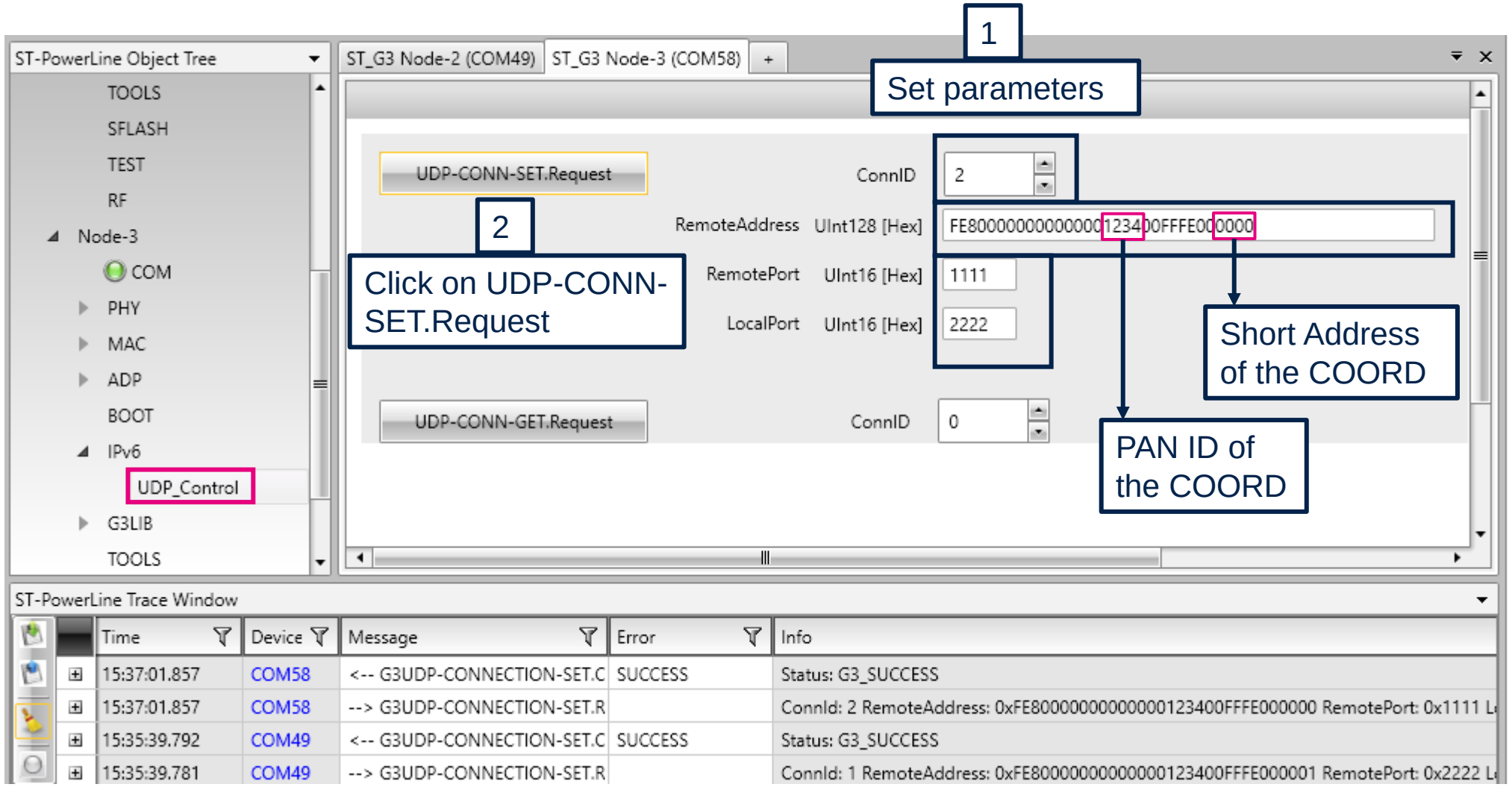
UDP settings: Coordinator

Coordinator node

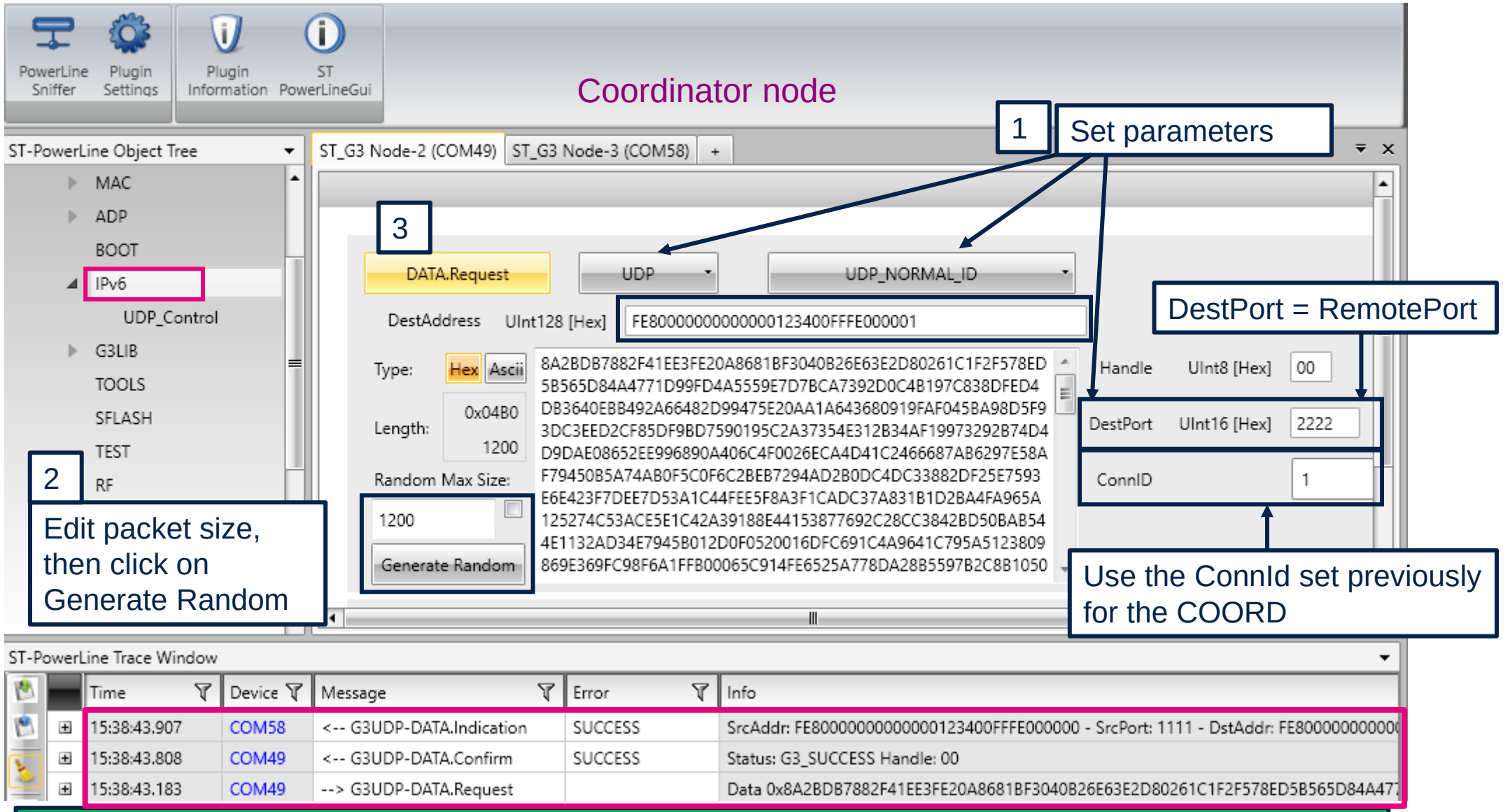


UDP settings: Device

Device node

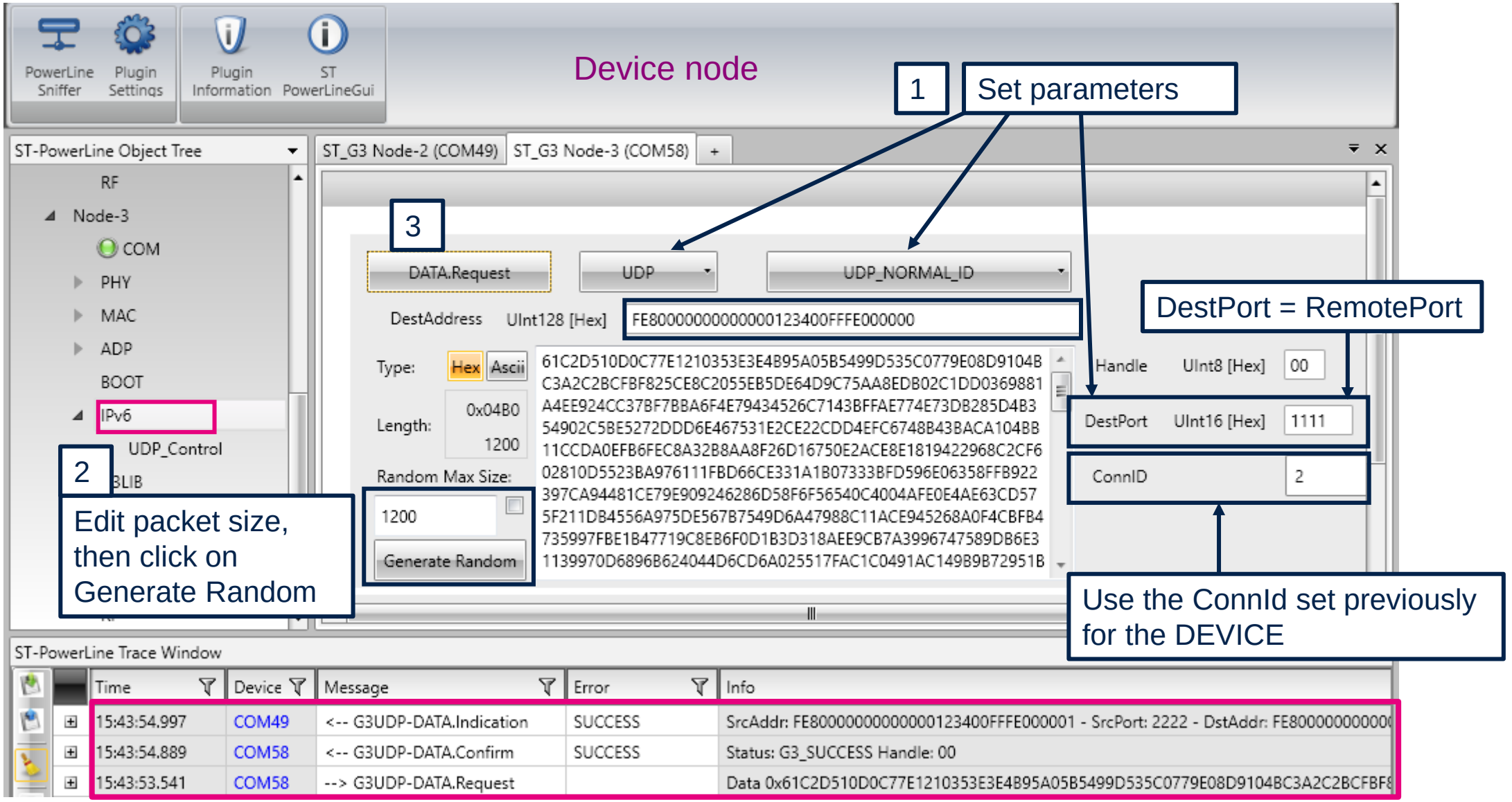


Coordinator → Device



G3 UDP Data Confirm to Coordinator (COM49) and G3 UDP Data Indication to Device (COM58)

Device → Coordinator



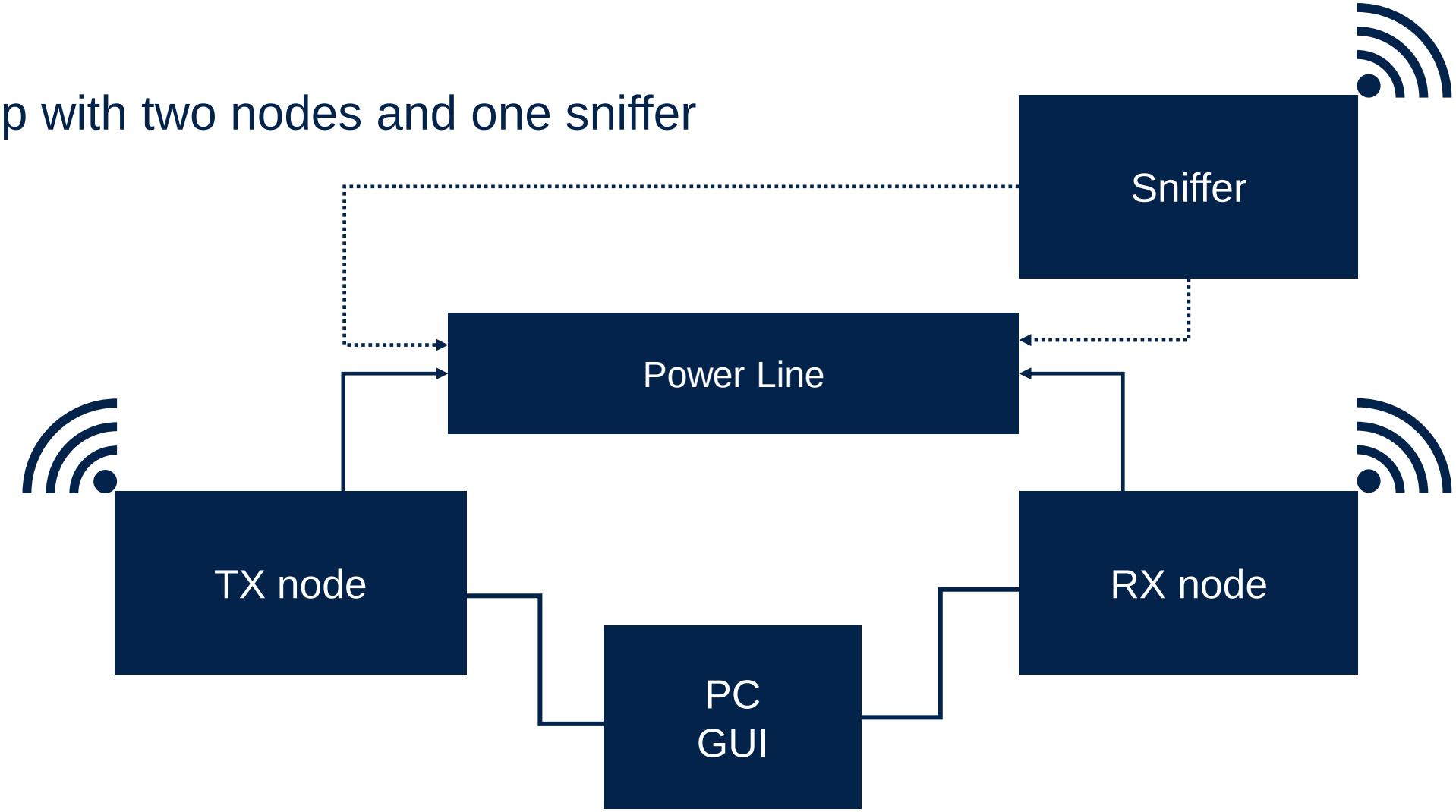
G3 UDP Data Confirm to Device (COM58) and G3 UDP Data Indication to Coordinator (COM49)

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Sniffer mode

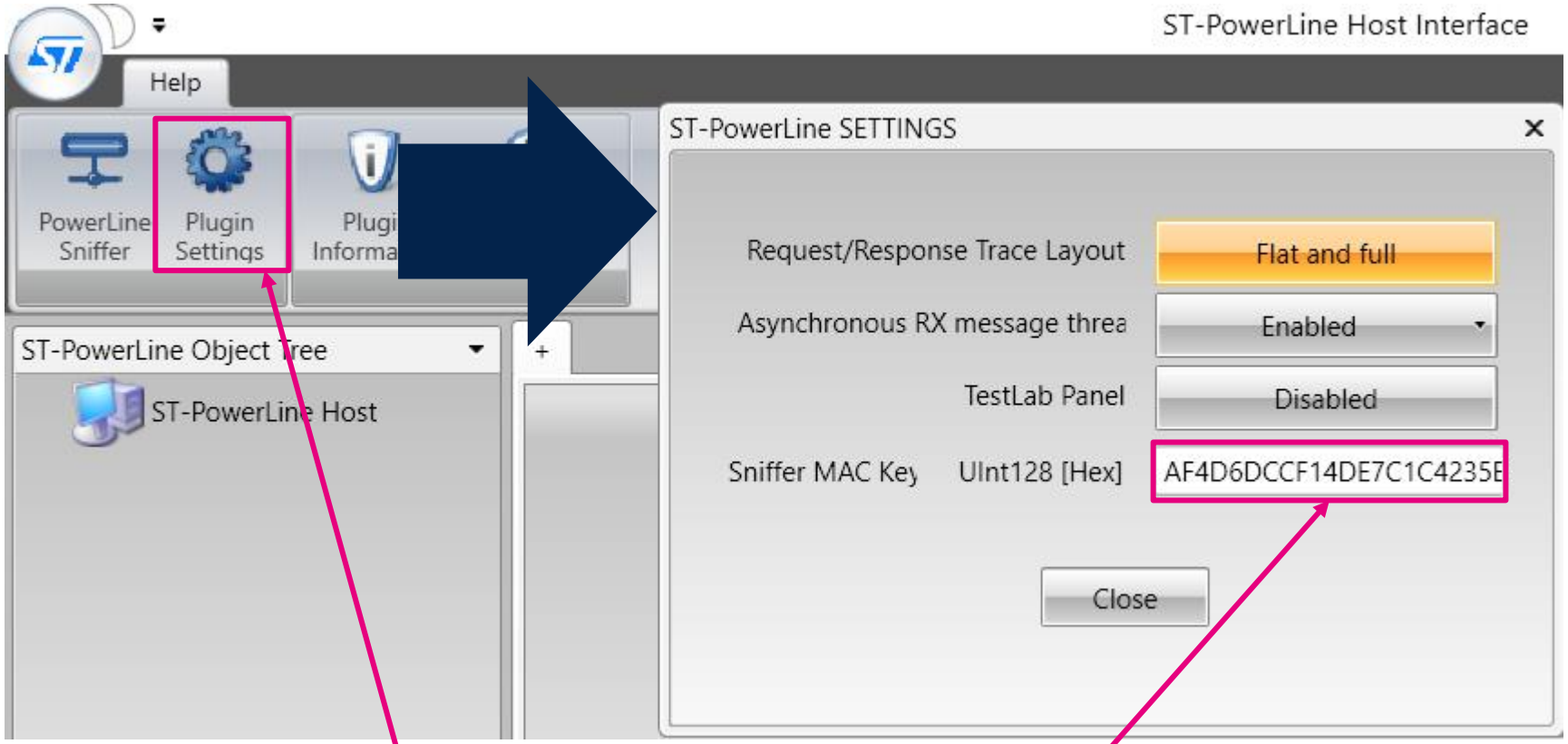
Communication Test setup with sniffer

- Setup with two nodes and one sniffer



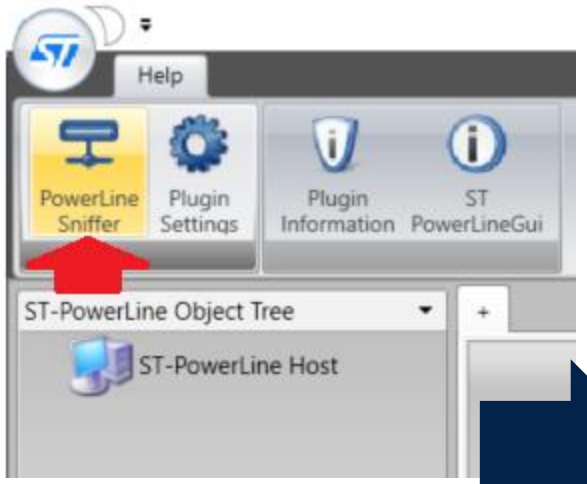
Note: the sniffer can be connected on the TX side or the RX side of the power line

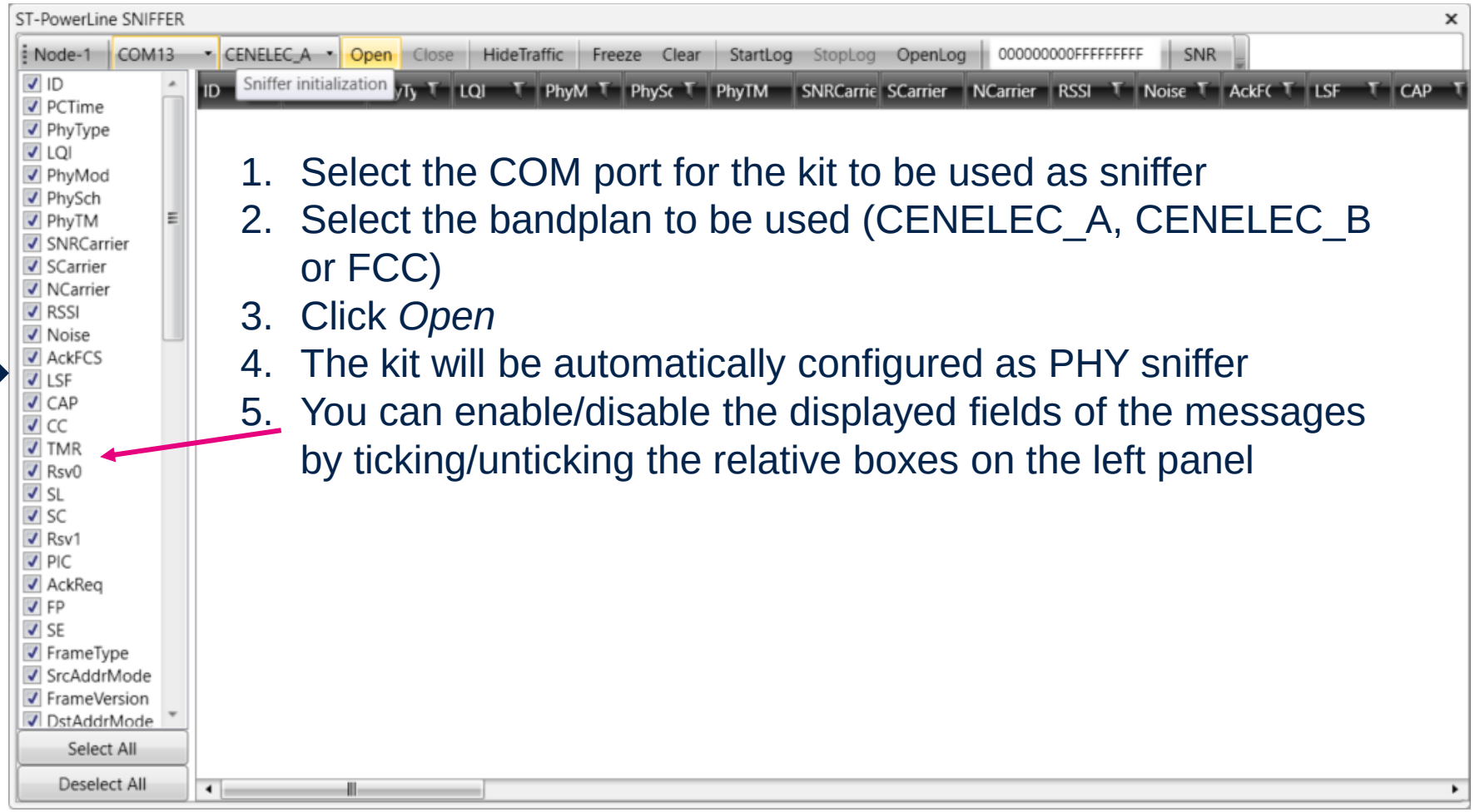
Sniffer mode configuration



1. Click on Plugin Settings
2. Modify the Sniffer MAC Key, using the GMK of the network to sniff, if different from the default value (AF4D6DCCF14DE7C1C4235E6FEF6C151F)
3. Click on Close

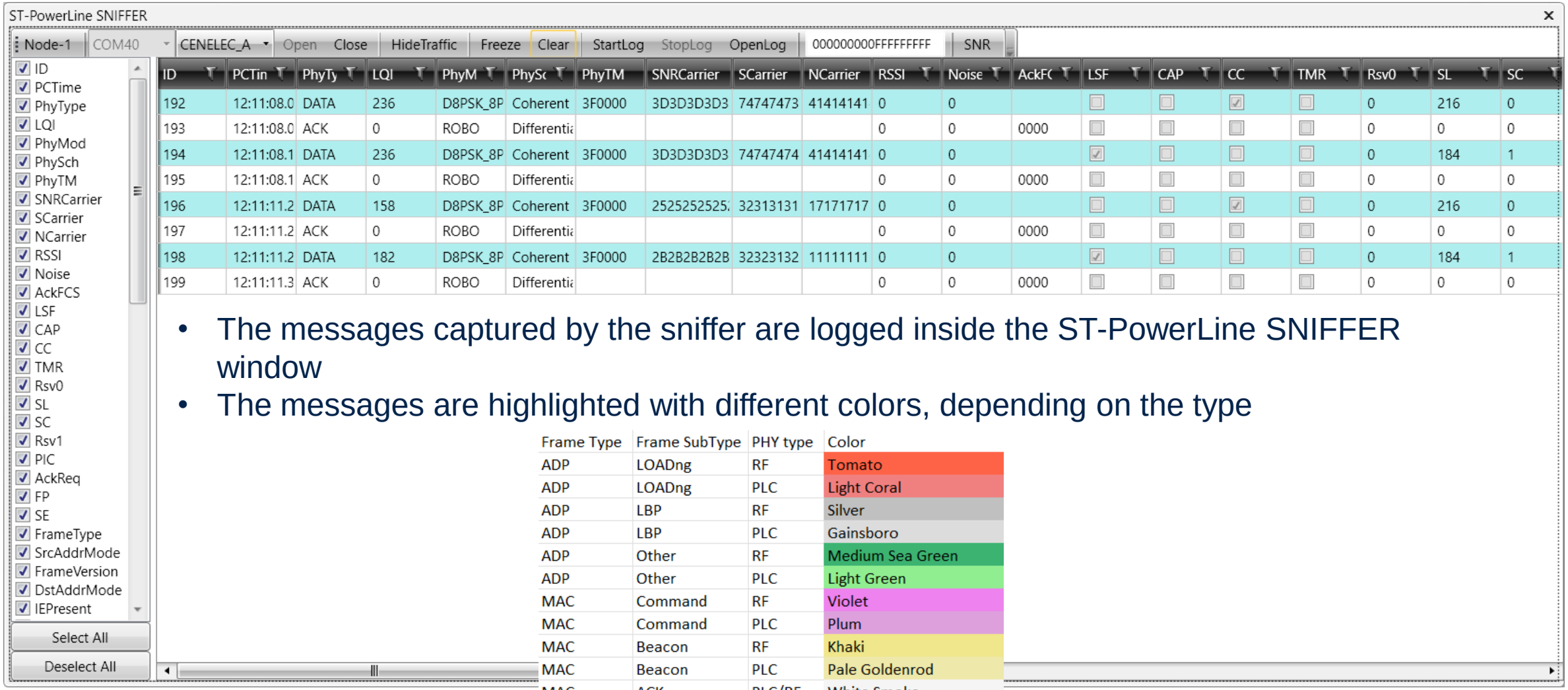
Sniffer mode activation





1. Select the COM port for the kit to be used as sniffer
2. Select the bandplan to be used (CENELEC_A, CENELEC_B or FCC)
3. Click *Open*
4. The kit will be automatically configured as PHY sniffer
5. You can enable/disable the displayed fields of the messages by ticking/unticking the relative boxes on the left panel

Sniffer mode traffic example



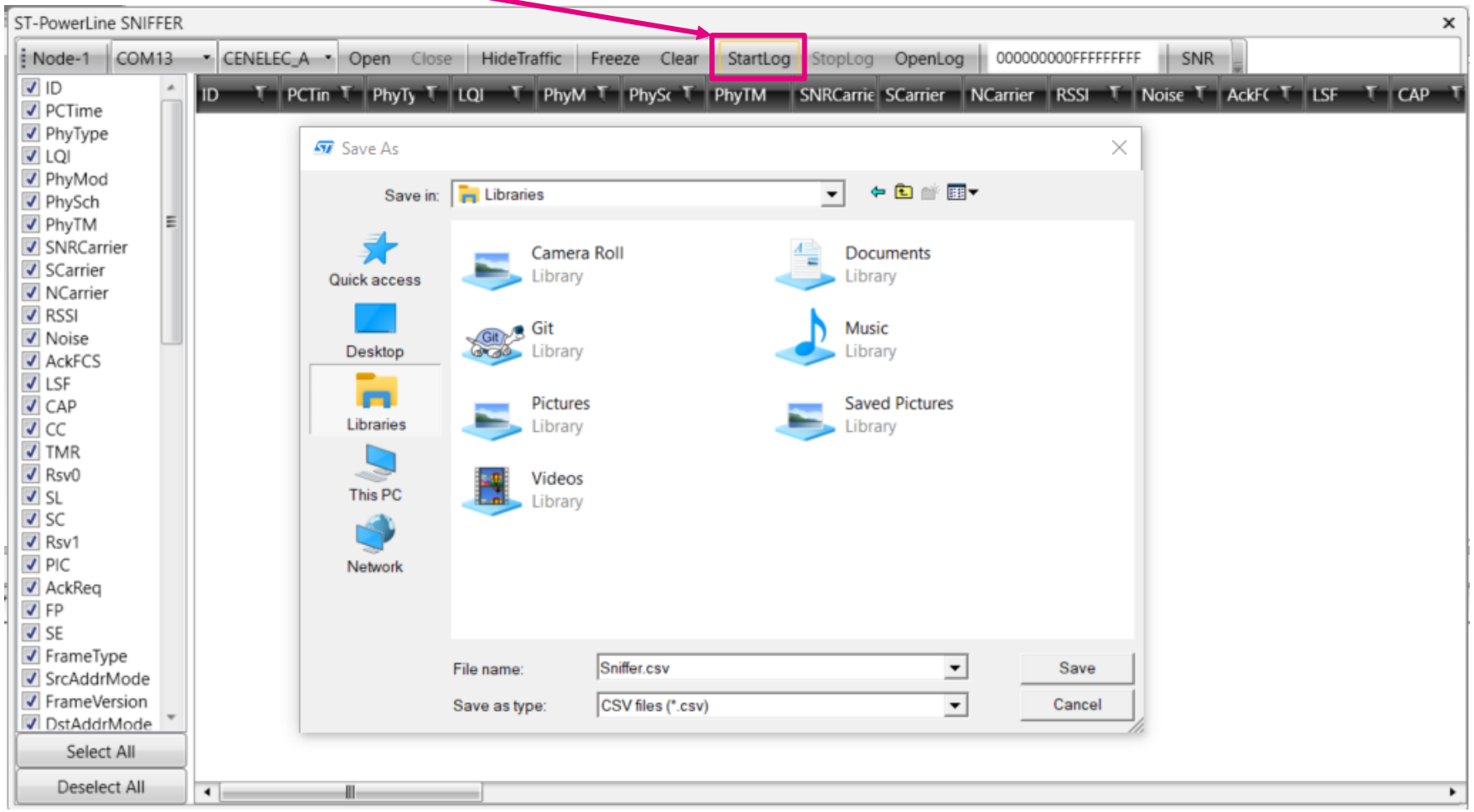
ID	PCTin	PhyTy	LQI	PhyM	PhySc	PhyTM	SNRCarrier	SCarrier	NCarrier	RSSI	Noise	AckFC	LSF	CAP	CC	TMR	Rsv0	SL	SC
192	12:11:08.0	DATA	236	D8PSK_8P	Coherent	3F0000	3D3D3D3D3	74747473	41414141	0	0		☐	☐	☑	☐	0	216	0
193	12:11:08.0	ACK	0	ROBO	Differentia					0	0	0000	☐	☐	☐	☐	0	0	0
194	12:11:08.1	DATA	236	D8PSK_8P	Coherent	3F0000	3D3D3D3D3	74747474	41414141	0	0		☑	☐	☐	☐	0	184	1
195	12:11:08.1	ACK	0	ROBO	Differentia					0	0	0000	☐	☐	☐	☐	0	0	0
196	12:11:11.2	DATA	158	D8PSK_8P	Coherent	3F0000	2525252525	32313131	17171717	0	0		☐	☐	☑	☐	0	216	0
197	12:11:11.2	ACK	0	ROBO	Differentia					0	0	0000	☐	☐	☐	☐	0	0	0
198	12:11:11.2	DATA	182	D8PSK_8P	Coherent	3F0000	2B2B2B2B2B	32323132	11111111	0	0		☑	☐	☐	☐	0	184	1
199	12:11:11.3	ACK	0	ROBO	Differentia					0	0	0000	☐	☐	☐	☐	0	0	0

- The messages captured by the sniffer are logged inside the ST-PowerLine SNIFFER window
- The messages are highlighted with different colors, depending on the type

Frame Type	Frame SubType	PHY type	Color
ADP	LOADng	RF	Tomato
ADP	LOADng	PLC	Light Coral
ADP	LBP	RF	Silver
ADP	LBP	PLC	Gainsboro
ADP	Other	RF	Medium Sea Green
ADP	Other	PLC	Light Green
MAC	Command	RF	Violet
MAC	Command	PLC	Plum
MAC	Beacon	RF	Khaki
MAC	Beacon	PLC	Pale Goldenrod
MAC	ACK	PLC/RF	White Smoke
MAC	Other	RF	Light Sky Blue
MAC	Other	PLC	Pale Turquoise
PHY	All	PLC/RF	White

Sniffer mode log

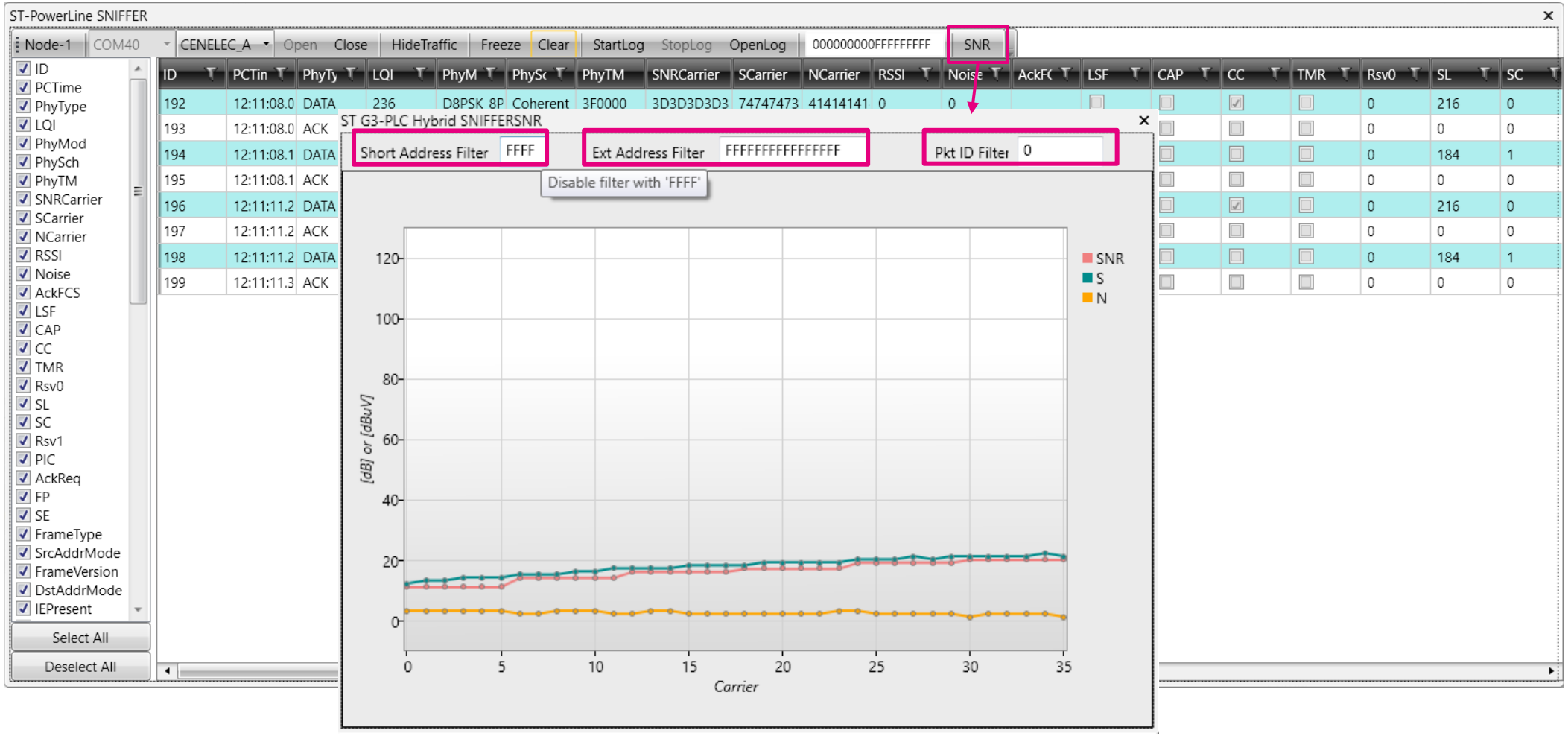
By clicking on *StartLog*, the log of the incoming traffic will be saved into a .csv file



Note: click on *StopLog* to close the .csv file and end the logging activity

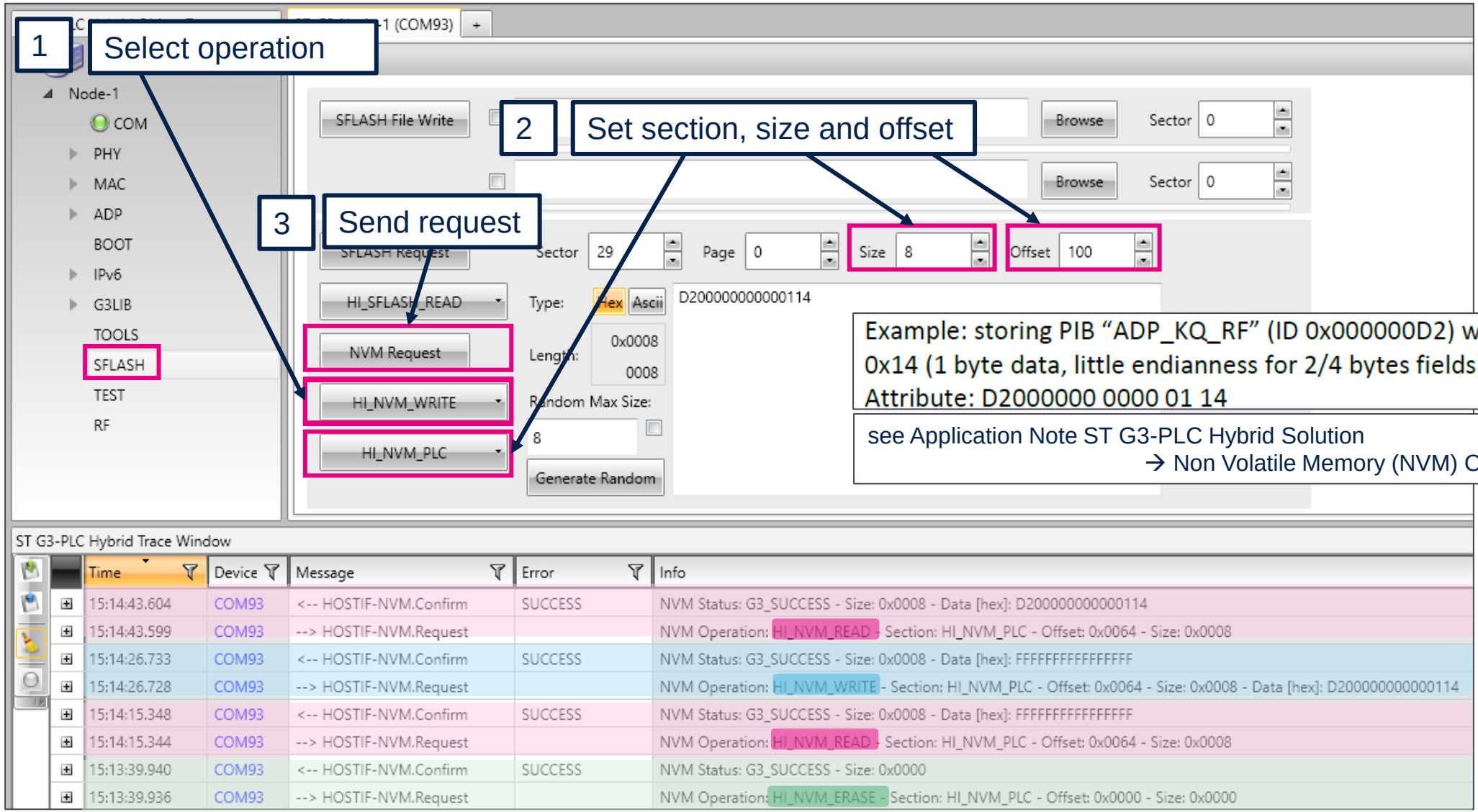
Sniffer mode SNR panel

By clicking on *SNR*, the SNR panel is opened.
The panel shows the SNR, S, N data extracted from the incoming G3PHY-DATA.Indication messages
Messages can be filterd by ShortAddr, ExtendedAddr or PktID



NVM operations

NVM operations



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Thank you

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